
When Does a Speech Model Know to Hand Off? Reading Competence from Hidden States for Real-Time Escalation

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Abstract

Real-time speech systems answer most queries with a local model, but some queries require a stronger expert. Selective escalation is well studied for text, yet speech adds a deployment question: does a speech model expose a reliable competence signal before it commits to a response, and does that signal survive distribution and modality shift? We compare decode-time uncertainty, verbalized self-assessment, and linear probes across layers and matched model controls, using leave-one-query-pool-out evaluation, matched text and speech inputs, and end-to-end escalation loops. On MiniCPM-o 4.5, an in-distribution final-layer probe scores AUC 0.822 but falls to 0.372 on held-out math with text input; the same last-token read at layer 22 reaches 0.931, while the cause of the late-layer failure remains unresolved. A text-calibrated layer-22 probe reads speech-input states at AUC 0.855, and a single-position end-of-turn study reaches 0.887 in ~ 30 ms, before the first output token. The live system uses the same layer with a separately calibrated three-position probe. In the primary turn-based, audio-input/text-output evaluation, selective escalation raises deployed-channel answer accuracy from 42.2% to 63.3% at 57% escalation and beats matched-rate random escalation by 7.2–9.5 points; a separate TTS-on replay leaves gate decisions unchanged when speech synthesis is enabled. With its weights, features, and layer fixed, the gate also improves all five external speech-QA benchmarks over always-local operation, although it does not beat matched-rate random escalation on every pool. These results identify a mid-network representation, rather than the conventional final-layer read, as a practical signal for handoff before response commitment.

1 Introduction

Real-time spoken interaction leaves only a few hundred milliseconds between turns [Stivers et al., 2009]. Systems that meet this deadline often rely on a compact interaction model, but no compact model can answer every long-tail factual, technical, or multi-step question. A stronger cloud model can handle some of these turns, provided the system knows when to call it. In a text interface, this is a standard routing problem. In a speech interface, the decision must also arrive before the local talker commits to a response; otherwise the system must delay the expert call or retract an answer already under way. We keep the local speech model as the talker so that the audio context and voice remain in one session. This setting raises the central question of this paper: *does a duplex-trained speech model expose its own likely failure early enough to hand the turn to a stronger model?*

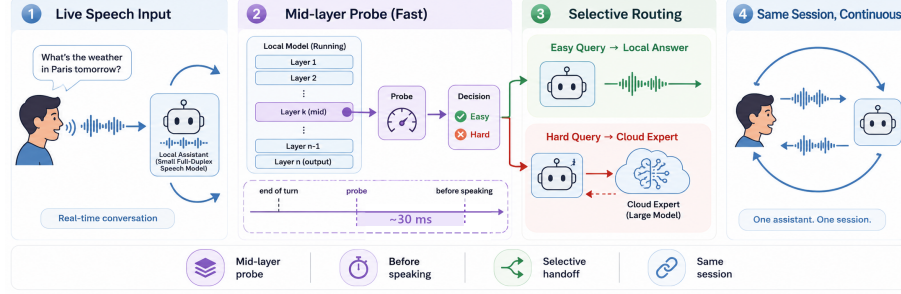


Figure 1: **Primary turn-based system.** (1) Speech reaches a duplex-trained local model. (2) At the end of the user turn, a mid-network (L22) linear read scores the turn in ~ 30 ms, before the first output token. (3) Lower-risk turns stay local; higher-risk turns are sent to a cloud expert. (4) The local model remains the talker and relays the returned answer. Deployed-channel accuracy rises from 42.2% to 63.3% at 57% escalation in the audio-input/text-output sweep; a TTS-on replay validates the speech path separately.

Two lines of work provide only part of the answer. Text-language-model studies estimate correctness from verbal self-assessment, output uncertainty, or hidden states [Kadavath et al., 2022, Azaria and Mitchell, 2023, Burns et al., 2023, Kossen et al., 2024]; text routers then use query-side or pre-generation features to choose a model [Ong et al., 2024, Varshney et al., 2026, Lugoloobi et al., 2026]. Speech-model research instead focuses on when a duplex model should listen, speak, or invoke a trained thinking policy [Wang et al., 2024a, Fang et al., 2026, Huang et al., 2026]. These results do not establish whether a frozen duplex-trained checkpoint retains a competence read that survives both distribution shift and speech input, or whether such a read is available before response commitment.

Three linked conditions make this gap operational. First, a useful score must distinguish individual failures rather than memorize the type of query seen during calibration. Second, a score calibrated on inexpensive text data must remain meaningful when the same content arrives as audio. Third, it must be available early enough to launch the expert and must improve which turns are escalated at a fixed call budget. A signal that works only in distribution, only on text, or only after a full draft cannot support real-time handoff.

We test these conditions on MiniCPM-o 4.5 [Cui et al., 2026], a 9B omni model with full-duplex training. We compare decode-time uncertainty, verbalized self-assessment ($p(\text{True})$), and linear hidden-state probes across layers and positions. Matched controls span raw backbones, streaming-omni and audio-only fine-tunes, a frozen-backbone audio adapter, and a second duplex generation. The controlled study uses judged failure labels on 600 public-benchmark queries in five frozen pools, with leave-one-pool-out (LOPO) transfer as the primary robustness test and matched text and speech inputs for modality transfer. We then measure when the score becomes available, insert it into a complete escalation loop, and freeze its layer and features for evaluation on public speech benchmarks. Model checkpoints remain frozen; only logistic probes are fit on stored activations. Throughout, *duplex-trained* describes the weights. The primary live evaluation uses a turn-based, end-of-turn decision with audio input and text output. A TTS-on replay measures speech onset, while separate experiments test concurrent and native full-duplex regimes (Appendices H and O).

The results form one evidence chain:

1. **Transfer changes where competence can be read.** The conventional final-layer probe scores AUC 0.822 in distribution but falls to 0.372 on held-out math with text input. On the same LOPO test, the last-token read at L22 reaches 0.931. Matched comparisons localize the late-layer failure to the duplex checkpoints we test, but direct controls do not identify its mechanism (Appendix D).

66 2. **The mid-layer read survives speech and arrives before response commitment.** A text-
67 calibrated L22 probe reads speech-input states at AUC 0.855, with the same mid-layer band
68 reproduced on real human recordings. In a single-position read-point study, the end-of-
69 turn read reaches AUC 0.887 in ~ 30 ms, before the first output token. A separate TTS-on
70 replay leaves the gate decisions unchanged and confirms that the read precedes synthesized
71 speech.

72 3. **The early read improves selective escalation.** A separately calibrated three-position
73 probe at the same layer drives the primary audio-input/text-output live loop. Deployed-
74 channel accuracy rises from 42.2% to 63.3% at 57% escalation and exceeds matched-rate
75 random escalation by 7.2–9.5 points. With its layer and features frozen, the gate improves
76 all five external speech-QA benchmarks over always-local operation, although selectivity
77 over random is limited on some pools. The native duplex regime also works after regime-
78 specific recalibration.

79 The central result is therefore narrower than a mechanism claim: the model does not explain why it
80 will fail, but its likely failure becomes more reliably readable mid-network than at the conventional
81 output interface, early enough to support a live handoff.

82 2 Related Work

83 **Pre-answer competence estimation.** Text-language-model correctness can be estimated from ver-
84 bal self-evaluation [Kadavath et al., 2022, Lin et al., 2022], output uncertainty, or hidden representa-
85 tions [Azaria and Mitchell, 2023, Burns et al., 2023, Marks and Tegmark, 2024]. Intermediate lay-
86 ers can carry more useful information than the output distribution [Orgad et al., 2025], and trained
87 hidden-state probes are strong factual-confidence estimators [Mahaut et al., 2024]. Semantic entropy
88 improves uncertainty estimation by comparing sampled answers [Kuhn et al., 2023, Farquhar et al.,
89 2024]; Semantic Entropy Probes amortize this signal into a pre-generation or post-hoc hidden-state
90 read [Kossen et al., 2024]. Attention-map probes and multimodal uncertainty fusion extend the same
91 broad idea [Chuang et al., 2024, Zhang et al., 2026a]. This work establishes that competence is read-
92 able in text models. It does not show where the read remains reliable after duplex speech training or
93 whether a probe calibrated on text transfers to speech-input states.

94 **Selective routing.** HybridLLM [Ding et al., 2024], RouteLLM [Ong et al., 2024], and query-side
95 routers [Xue et al., 2025, Hu et al., 2024, Li et al., 2026] choose a model for a complete text query
96 from query features or preference data. FrugalGPT organizes cascades that can score an earlier
97 model’s response [Chen et al., 2023], while AutoMix verifies a drafted answer before escalating
98 [Aggarwal et al., 2024]. Closest to our signal, recent methods read pre-generation activations or
99 fit linear success probes on frozen states [Varshney et al., 2026, Lugoloobi and Russell, 2025, Lu-
100 goloobi et al., 2026]. A second family trains the decision into the generator: Toolformer and Self-
101 RAG learn tool or reflection tokens [Schick et al., 2023, Asai et al., 2024]; Co-LLM and CITER
102 learn token-level deferral [Shen et al., 2024, Zheng et al., 2025]; and pause, thought, or think/no-
103 think tokens allocate more internal computation [Goyal et al., 2024, Zelikman et al., 2024, Yang
104 et al., 2025, DeepSeek-AI et al., 2025]. Learning-to-defer and early-exit methods make related de-
105 cisions within a model [Madras et al., 2018, Schuster et al., 2022]. Our contribution is not the linear
106 pre-generation probe by itself. We test whether this kind of read survives duplex training and audio
107 input, and use a deployment-time threshold without modifying the speech checkpoint.

108 **Duplex control and live handoff.** End-to-end spoken dialogue includes native duplex models
109 [Nguyen et al., 2023, Défossez et al., 2024, Zeng et al., 2024, Fang et al., 2026], streaming omni
110 models [Xu et al., 2025, Cui et al., 2026], frozen-backbone adapters [Wang et al., 2024b], and audio
111 fine-tunes [Chu et al., 2024]. Their control interfaces decide when to listen or speak through state,
112 idle, synchronization, or interruption tokens [Wang et al., 2024a, Zhang et al., 2024, Veluri et al.,
113 2024, Ma et al., 2024, Fu et al., 2024, Yu et al., 2024], structured action channels [Zhang et al.,
114 2026b], or small classifier heads [Liao et al., 2025, Chen et al., 2025]. MiniCPM-o 4.5 likewise

115 predicts a listen/speak state before content generation [Cui et al., 2026]. These policies answer
116 *when* the model should speak, not whether its content is likely to be correct.

117 Recent systems also connect a duplex interface to a stronger resource. MoshiRAG triggers asyn-
118 chronous retrieval from the model’s emerging output and returns knowledge to the local model
119 [Chien et al., 2026]. DuplexOmni uses a trained think-control token to hand work to a separate
120 reasoning layer [Huang et al., 2026]. These systems demonstrate that a live handoff is possible, but
121 they do not evaluate a competence score read from a frozen duplex checkpoint before generation.

122 **Positioning.** Prior work therefore establishes three ingredients separately: competence can be read
123 from text models, text queries can be routed, and duplex models can coordinate speaking or external
124 computation. To our knowledge, the reviewed literature does not establish their conjunction:
125 whether a frozen duplex-trained speech model exposes a pre-answer competence signal that sur-
126 vives unseen query pools and text-to-speech transfer, and whether that signal arrives early enough
127 to improve live selective escalation. We evaluate these conditions directly. The claims are about
128 readout robustness and system utility; the training mechanism behind the late-layer failure remains
129 open.

130 3 Experimental Setup

131 **Target model and matched-pair controls.** The system target is **MiniCPM-o 4.5** (9B; omni +
132 full-duplex fine-tune of Qwen3-8B), bf16, greedy decoding, seed 42. To attribute effects to the *kind*
133 of fine-tune rather than the backbone, we assemble ten models spanning raw backbones, a second du-
134 plex generation (MiniCPM-o 2.6), a streaming-omni fine-tune (Qwen2.5-Omni-7B), an audio-only
135 fine-tune (Qwen2-Audio-7B), and a frozen-backbone adapter model (Freeze-Omni, LLM weights
136 byte-identical to its raw control); the full roster is Table 2 (appendix). The comparison statistic is
137 always the within-pair difference.

138 **Query pool, labels, and audio arm.** 600 queries in five pools, frozen before any gate experi-
139 ment (360 calibration / 240 test): *trap* = SimpleQA [Wei et al., 2024] (the target fails 100%), *hard-*
140 *knowledge* = MMLU-Pro [Wang et al., 2024c], *easy-fact* = TriviaQA [Joshi et al., 2017], *easy-chat*
141 = Dolly/Alpaca [Conover et al., 2023, Taori et al., 2023], *hard-math* = GSM8K tail + MATH-500
142 [Cobbe et al., 2021, Hendrycks et al., 2021, Lightman et al., 2024]; public benchmarks only, no
143 LLM-generated queries. Adequacy labels come from an LLM judge [Zheng et al., 2023] with struc-
144 tured outputs, blind to the gate and validated by a stronger re-judge (agreement 96.2% text / 94.5%
145 audio; every headline number moves ≤ 1 point under the stronger labels); the escalation expert is
146 gpt-5.5 at low reasoning effort. The audio arm renders the same frozen pool to speech (single-
147 voice TTS, content unchanged — only modality varies), following Spoken-SQuAD / VoiceBench
148 practice [Li et al., 2018, Chen et al., 2024]; audio-side findings are additionally validated on real
149 human recordings (SD-QA, Faisal et al., 2021) and on the external benchmarks’ official audio (§7).
150 Judge prompts, pins, and spend: Appendix M–N.

151 **Candidate competence signals.** All model checkpoints remain frozen. We compare (i) decode
152 scalars: max token entropy or margin over the first k decode steps; (ii) linear probes: logistic re-
153 gression on hidden states, with layer and position selected on calibration data; and (iii) verbalized
154 self-evaluation [Kadavath et al., 2022]: $p(\text{True})$ -pre asks “would you answer this correctly?” before
155 answering, $p(\text{True})$ -post asks about a drafted answer; $P(\text{Yes})$ is read from the first token’s distri-
156 bution. The representation analyses use a single-position, 4096-d L22 probe fit on the 360-query
157 calibration split. The live system uses the same layer plus two pooled positions, fit on a wider 2,310-
158 query calibration mix that is disjoint from the external benchmarks. Concurrent and native serving
159 change the read point, so those validations refit the same feature set in regime; Appendix B records
160 the full calibration ledger.

Evaluation logic. In-distribution AUC is diagnostic, not the main robustness claim, because query-pool identity is predictive in the calibration mix. We therefore use LOPO AUC to test distribution transfer and text-trained/audio-scored AUC to test modality transfer. For the live system, the key comparison is against matched-rate random escalation: an improvement over always-local operation can come from the expert alone, while an improvement over random at the same call rate measures the value of the gate’s ranking.

4 A Transferable Competence Read Survives Mid-Network

4.1 In-distribution accuracy hides a query-type shortcut

Table 3 compares the candidate signals on MiniCPM-o 4.5. In distribution the final-layer probe looks healthy at AUC 0.822 — but that number is mostly a *query-type* shortcut. The same hidden state all but names which pool a query came from, a pool-identity oracle alone recovers most of the probe’s in-distribution margin. Appending true pool labels to the probe’s features adds nothing because pool identity is already encoded. We therefore treat leave-one-pool-out (LOPO) transfer as the primary metric. With text input, the final-layer read falls to AUC 0.372 on held-out math under LOPO. When the trap pool is held out, its mean failure score is 0.232 even though the model fails every query in the pool. A query-surface router shows the same dependence on pool structure: it does not exceed the pool oracle in distribution and also degrades under LOPO, with little improvement from substantially more data (Appendices B, E). These results show why in-distribution routing accuracy alone cannot establish instance-level transfer.

Verbalized self-assessment transfers differently. Asked *before* answering, $p(\text{True})$ catches exactly the trap pool the probe misses, scoring it 0.945. But timing is everything: asked *after* drafting, the model endorses its own confident-wrong answer and the signal inverts. Post-draft $p(\text{True})$ is the strongest single readout in distribution, at 0.899, but it no longer provides a pre-answer routing signal.

4.2 Late-layer brittleness appears on text input in duplex checkpoints

The matched-pair pattern ties the failure to the duplex checkpoints rather than to the backbone alone. The raw Qwen backbone, subsampled to match MiniCPM-o 4.5’s per-pool failure rates, holds LOPO math at 0.962–0.968 across five seeds where the duplex checkpoint scores 0.372. The Qwen2.5 family shows the same ordering: the raw backbone scores 0.809 (0.825 after failure-rate matching), the streaming-omni model scores 0.745, and the duplex MiniCPM-o 2.6 scores 0.538. The Qwen2-Audio comparison is not attribution evidence because the audio fine-tune also changes task capability and the failure distribution substantially (Appendix B). A layer $\times$ position sweep further localizes the difference (Figure 2): it is concentrated at the last-token position in late layers on text input. Mean-pooled reads remain usable at the same depth but trail the mid-layer peak by 0.13–0.15. At L22, the last-token probe reaches **0.931** on the same LOPO math test, close to the raw backbone.

The failure is input-specific rather than universal: with audio calibration and audio input, the final-layer probe reaches 0.936 on LOPO math. We use L22 because one read point must remain robust for shifted text queries and support text-to-speech transfer, not because the final-layer signal is absent on the model’s native audio input.

The sweep establishes *where* the transferable read survives, not *why* it weakens late. Direct tests do not support the simple hypothesis that turn-control state takes over the last-token representation: the layer at which performance drops does not move in speak mode, and the inverting probe direction aligns with neither the control vocabulary nor an explicit listen/speak direction. A decomposition further shows that a distributed transferable component weakens over the final layers of the two duplex checkpoints but not their raw backbones. These observations constrain the explanation but do not identify a training objective or causal mechanism; the full audit is in Appendix D.



Figure 2: The core representational result: LOPO math AUC by layer and position. On the duplex model the standard (final-layer, last-token) read inverts on text input while mid-network layers hold ≥ 0.9 ; raw backbones show no cliff. The deployed gate reads L22.

208 The mid-network pattern also appears outside the MiniCPM family. On NemotronLabs-VoiceChat-
 209 11B, a hybrid-Mamba duplex model, the same recipe peaks at L30–34 of 56 and transfers to the
 210 external pools at AUC 0.70–0.79 with one quarter of the deployed probe’s calibration data (Ap-
 211 pendix C).

212 4.3 Text-calibrated reads transfer to speech in end-to-end models

213 Probes trained on text hidden states and scored on speech-input hidden states locate where the repre-
 214 sentation becomes modality-shared: early layers are modality-specific (0.54–0.60), and from $\sim 50\%$
 215 depth text→audio transfer plateaus at 0.82–0.87 (0.855 at the deployed L22). This replicates on
 216 MiniCPM-o 2.6 and cross-family on Qwen2.5-Omni. The decisive control is **Freeze-Omni**: identi-
 217 cal frozen LLM weights with audio attached by an adapter — and cross-modal transfer remains at
 218 0.52–0.60. This comparison supports a narrow operational rule: text calibration transfers in the end-
 219 to-end speech models tested here, but not in the adapter-attached control. Verbalized introspection
 220 shows the complementary failure: on audio input, pre-answer $p(\text{True})$ collapses on both MiniCPM
 221 duplex generations while both non-duplex controls stay intact; controlled arms show the verbal self-
 222 check is sensitive to the text-token pathway — duplicating the question as text restores it — while
 223 a linear probe remains usable on the same forward pass (Appendix B). The restored self-check is
 224 additive with the probe: stacking the two lifts the internal test AUC 0.836→0.862 (CI excludes zero)
 225 at the cost of one short prefill per turn (Appendix Table 7). The mid-layer band also replicates on
 226 200 real human recordings from SD-QA, so the result is not limited to the TTS rendering. The live
 227 system keeps L22 and adds two pooled positions to form the deployed probe (Appendix B).

228 5 The Read Arrives Before Response Commitment

229 **Scoring latency.** The gate is a linear read on L22 states the forward pass already computes —
 230 a 4096-d dot product per position, over one position (mechanistic probe) or three (the deployed
 231 probe adds two running means at no added scoring latency; Appendix B). On an H100, measured
 232 from prefill start with CUDA synchronization, the score is available at P50 in **20 ms for text and**
 233 **45 ms for audio** (P95 25/104 ms). Time to first token on the same benchmark is 36/68 ms. The
 234 measurement excludes network time, but it establishes the ordering needed by the system: the cloud
 235 request can start before local decoding commits to an answer. Waiting for decoded tokens does not
 236 improve the read in our data. Max entropy over four tokens reaches AUC 0.696, and early-decode
 237 hidden states reach 0.776, compared with 0.828 at the prompt; waiting also postpones the expert
 238 request (Appendix F).

239 **From score to escalation budget.** On the frozen test split ($n=240$), always-local accuracy is
 240 58.8% and always-expert accuracy is 91.7%. Thresholding the probe beats matched-rate random
 241 escalation at every operating point (Figure 7, appendix). At a 33% escalation budget, it reaches
 242 **77.9–78.7%**, recovering about 58% of the local-to-expert gap with one third of the expert calls. In

distribution, the gate is statistically tied with a pool-identity oracle, as the shortcut audit predicts. Its advantage appears under transfer, where pool identity stops being sufficient (§4.1).

The gate is fit to *local failure*, not expert benefit: a benefit label would specialize the ranking to the current expert. The frozen scores remain useful under the stricter benefit label, but pools with little expert headroom remain a system-level limit (Appendix I). Comparisons with the model’s built-in reasoning mode are in Appendix F.

6 Live End-of-Turn Escalation

Primary protocol. The protocol is turn-based: audio streams in 1 s chunks, but the talker speaks only after the user’s turn ends and the model’s native listen/speak head is not engaged. This protocol isolates competence handoff from turn-taking; the native-head validation is reported separately below. Scores from intermediate chunks are weaker than the offline read (best summary AUC 0.764) and can miss information in the last chunk. We therefore make one **end-of-turn read**: after the final audio chunk, the system prefills the assistant turn and reads L22 at its start. In the 360-query read-point study, the single-position probe reaches **AUC 0.887** in ~ 30 ms, compared with 0.843 for the offline reference, and arrives before the first output token. This experiment establishes the read point. The accuracy sweeps below use the deployed three-position probe fit on the wider 2,310-query calibration mix (frozen-test AUC 0.879); tier thresholds are quantiles of its score.

If the turn is escalated, the local talker first produces a short stall phrase. Its own transcription of the input is sent to gpt-5.5, and the returned answer is injected as context for the local model to relay. The expert request runs asynchronously. Partial-query prefetch, expert reasoning effort, and injection controls are reported in Appendix G.

The live result. We ran 240 frozen test queries $\times$ four escalation tiers, each a complete turn-based loop with audio input. The primary accuracy sweep retains the model’s output as text; a separate TTS-on run of the balanced tier verifies that enabling speech synthesis leaves all 84/240 gate decisions unchanged and measures speech onset. We score *deployed-channel answer accuracy*: correctness of the relayed content after the system path. Headline numbers use a pre-registered input filter that removes 21 formula-LaTeX queries and one broken recording, because the speech rendering destroys the symbolic input before the model receives it. Accuracy rises monotonically, **42.2% $\rightarrow$ 52.8% $\rightarrow$ 59.2% $\rightarrow$ 63.3%** at 0/14/31/57% escalation, every tier’s gain significant (paired bootstrap; Figure 12). More importantly, it exceeds the matched-rate random reference at every tier, by 7.2–9.5 points in the deployed view and by 5.4–8.4 points in the channel-controlled view (Figure 12). A measured always-escalate arm reaches 66.5%. The small remaining gain from 57% to 100% escalation is not evidence that selection is perfect: the same sessions reach 92.2% when the expert receives gold text, showing that the model’s transcription uplink is the limiting channel at high escalation rates. Live accuracies have a replication floor of ± 2 –3 points; no claim rests on a smaller difference. Full tables, latency profiles, and boundary cases are in Appendix G.

Serving-regime checks. A talker-on replay leaves the frozen ranking nearly unchanged (AUC 0.871/0.856 versus 0.866 headless). A teacher-forced concurrent interleave is less stable and requires an in-regime refit; its external results remain exploratory (Appendix H). Under the model’s native full-duplex head, the read moves to the head’s own listen $\rightarrow$ speak commit point. Refitting the same features in that regime yields AUC 0.830 internally and 0.709 across the external QA pools. A native outcome re-mix beats matched-rate random on five of six pools. A full native live run on the internal 240 queries improves accuracy from 0.371 to 0.537 at 45% escalation, below the 0.596 re-mix prediction because relaying the returned answer is lossy. These checks support the ranking beyond the primary turn-based loop, but they also show that read points, probe weights, and thresholds must be calibrated for the serving regime (Appendix O).

Table 1: Transfer to five public speech-QA benchmarks. The top block is measured post-relay answer accuracy (%) from the audio-input/text-output live loop; the probe is fixed and thresholds use label-free per-pool quantiles. The bottom block applies the same recipe to NemotronLabs-VoiceChat-11B in an offline re-mix (Appendix C). Δ is the average gain over always-local. AlpacaEval, reported separately on its 1–5 scale, changes from 3.99 to 4.35 on MiniCPM but does not beat matched-rate random; the second family changes from 3.55 to 4.46 versus 4.23 for random. Full random references and gold-input ceilings are in Appendices I–J.

	TriviaQA	WebQ	Llama Q.	SD-QA	Reason. zh	avg	Δ
<i>live post-relay view (audio input, text output)</i>							
always-local	66.4	57.2	84.0	51.5	58.9	63.6	—
+ gate @15%	73.2	62.8	90.4	60.0	65.3	70.3	+6.7
+ gate @30%	80.0	68.0	90.4	72.0	71.3	76.3	+12.7
+ gate @50%	86.0	73.2	94.8	79.5	77.2	82.1	+18.5
always-escalate (measured)	88.0	80.8	93.6	88.5	83.7	86.9	+23.3
always-expert (gold-text ceiling)	96.8	86.4	92.8	93.0	87.1	91.2	+27.6
<i>second family: NemotronLabs-VoiceChat-11B (offline re-mix, official judges)</i>							
always-local	35.6	37.6	70.5	31.0	—	43.7	—
+ gate @15%	49.8	48.0	78.6	42.5	—	54.7	+11.0
+ gate @30%	61.5	58.8	83.8	54.5	—	64.6	+21.0
+ gate @50%	74.1	69.6	87.6	67.5	—	74.7	+31.0
matched random @50%	65.6	60.2	81.8	60.0	—	66.9	+23.2
always-expert (ceiling)	95.5	82.8	93.2	89.0	—	90.1	+46.4
probe AUC (frozen)	0.789	0.785	0.806	0.792	0.683	0.771	—
probe AUC (2nd family)	0.775	0.771	0.720	0.754	—	0.755	—
official offline	75.5	70.2	—	—	—	—	—

7 Frozen Transfer to Public Speech Benchmarks

The query-type audit makes external transfer necessary. We freeze the probe weights, feature set, and layer selected on the internal calibration mix, then run the live loop of §6 on six public speech benchmarks with audio input and text output: Speech TriviaQA, Speech Web Questions, Llama Questions, and Reasoning QA from OpenAudioBench [Li et al., 2025] (spoken renderings of Joshi et al., 2017, Berant et al., 2013, and the set of Nachmani et al., 2024), SD-QA (real dialectal human speech) [Faisal et al., 2021], and VoiceBench AlpacaEval [Chen et al., 2024, Taori et al., 2023]. All audio is the benchmarks’ official pre-rendered audio; no evaluation query is used to fit the probe. The only per-pool adjustment is *label-free*: tier thresholds are quantiles of that pool’s score distribution, requiring no correctness labels or judge calls. This controls the call budget but cannot improve ranking. A single absolute threshold also beats matched-rate random on four of five QA pools, but its fire rate varies from 2% to 48% (Appendix I). We use each benchmark’s official judge when available and verify that the local-only controls are on the published scale.

The frozen scores remain discriminative outside the calibration mix: AUC ranges from 0.683 to 0.806 across the five QA pools, with a mean of 0.771 (Table 1). The gate also improves deployed-channel accuracy over always-local operation on every QA benchmark and at every budget; at 50% escalation, mean accuracy rises from 63.6% to 82.1%. These gains establish system utility but do not by themselves establish selectivity, because random escalation also adds expert answers.

Llama Questions provides the cleanest fixed-budget test. It contains a single query type, the kept-local subset scores 97.6% while the escalated subset scores 69.6% ($z=6.46$), and expert answers improve the latter to 88.8% ($p<.0001$). At 50% escalation, the measured system reaches 94.8%, compared with 93.6% for always escalating. Matched-rate precision reaches 95% of its base-rate cap (Appendix J). Cross-lingual transfer also improves when the English calibration mix is widened: Chinese Reasoning QA rises from AUC 0.621 to 0.683 despite adding no Chinese calibration data.

The transfer results also expose the boundary between predicting local failure and creating expert benefit. SD-QA is a positive real-speech case and exceeds matched-rate random at all three budgets. Web Questions is less stable, with little gain over random at the low and high budgets; in its top-scoring 15%, many local failures are also missed by the expert (Appendix I). AlpacaEval is a clearer negative: its open-ended failures do not present the retrieval-failure pattern the probe reads. On the second duplex family, which has more expert headroom on these pools, the same recipe beats

matched-rate random on all five evaluated pools, including AlpacaEval ($p \leq .0004$). Thus external utility depends on both the quality of the competence ranking and whether the selected failures are repairable by the expert (Appendices C, I, and J).

8 Discussion

What the experiments establish. The three parts of the study answer different requirements of the same handoff decision. First, in-distribution probe accuracy is not enough: query type explains much of the apparent performance, and LOPO evaluation changes the preferred read from the final layer to the middle of the network. Second, this mid-layer read remains useful when text calibration is applied to speech-input states and is available before response commitment. Third, its ranking improves a live escalation loop over matched-rate random selection. External and native evaluations support the same conclusion, but also show that neither probe weights nor thresholds should be assumed to transfer across serving regimes without validation.

What the score represents. The probe is strongest on retrieval-like failures, where the model’s state reflects a missing answer before generation. It is weak on false premises and complementary to signals that appear during decoding (Appendix G). This construct boundary explains why failure AUC and system benefit are not identical. A correct ranking cannot help when the expert fails on the same turns, and an open-ended quality benchmark may not contain the factual-retrieval event the probe detects. The unstable Web Questions curve and the negative AlpacaEval result are therefore part of the finding, not exceptions to hide; SD-QA, by contrast, confirms selective gains on real human speech. Selective escalation depends on both readable local failure and expert headroom.

Calibration is part of deployment. The model weights stay frozen, but the gate is still a fitted component. Text-only calibration is sufficient for the controlled cross-modal result on end-to-end speech models. The deployed three-position probe uses a wider benchmark-disjoint calibration mix, and both concurrent and native read points require in-regime refits. The method avoids fine-tuning the talker and lets the operating budget move through a threshold, but it does not eliminate the need for representative calibration data.

Limitations. (i) One phrasing per $p(\text{True})$ variant; real-speech validation is scripted read speech in one dialect; formula-symbolic content is out of scope for the audio arm (any TTS render destroys it). (ii) The second-family replication covers mid-band readability and frozen-recipe transfer only — not matched-pair attribution, the live loop, or the concurrent regime. (iii) Labels come from one judge family (a stronger re-judge bounds deltas at ≤ 1 point), no human labels. (iv) One session per query per tier (± 2 – 3 point floor); under concurrent prefill the frozen read degrades (AUC .87 $\rightarrow$.76) and an in-regime refit recovers .82 — calibration must follow the regime, and scale helps only partly (a full 2,310-row in-regime refit lifts external mean AUC .64 $\rightarrow$.69, still short of turn-based .77); that concurrent loop is exploratory (teacher-forced carrier, random-clearance internal but spotty externally; Appendix H). The deployed *native* full-duplex regime lands strictly between the two (in-regime AUC .83 internal / .71 external mean; Appendix O); its internal full live run is limited by a lossy relay channel. (v) Layer and features were chosen on internal calibration-split cross-validation and confirmed on held-out pools, not by nested selection inside each held-out pool (Appendix B). (vi) The mechanism behind the late-layer decay on text input is unresolved (Appendix D). (vii) The primary evaluation uses scripted, single-turn questions. Disfluent correction, chained tool calls, and multi-turn grounding remain open [Lin et al., 2026]; in the native test, a new question during the expert wait derails the pending relay in 7/10 trials, so production systems must cancel or replace an in-flight request when the user moves on.

Conclusion. In the duplex speech models studied here, the conventional final-layer read is not the most reliable place to estimate answer competence when text queries shift across pools. A mid-network read retains that query-pool transfer and remains usable when calibrated on text and scored on speech-input states. It also becomes available before response commitment and improves which

turns a live system escalates. The result supports a practical handoff mechanism while leaving the cause of the text-input late-layer failure open.

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573 A Model roster

Table 2: Model roster. “Pair” groups a fine-tune with its raw backbone control; the comparison statistic is always the within-pair difference.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	duplex replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio-FT control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control
Mistral-7B-v0.3	raw	—	backbone diversity
GLM4-9B-chat	raw	—	backbone diversity

574 B Extended signal and readout analyses

Table 3: Signals on MiniCPM-o 4.5, judged failure labels. In-dist = calibration-split AUC (probes 5-fold out-of-fold); LOPO math = leave-one-pool-out AUC on held-out math; audio = the same readout under speech input (final/mid-layer probe: audio-input LOPO math / text-only calibration read on audio states). Full per-signal detail: Appendix B.

signal	in-dist	LOPO math	audio input
max entropy@4	0.696	≈chance on raw backbones	—
$p(\text{True})$ -pre	0.807	trap 0.945 (probe: 0.232)	collapses
$p(\text{True})$ -post (full draft)	0.899	—	degrades
final-layer probe	0.822	0.372 (inverts)	0.936 (intact)
mid-layer (L22) probe	0.866	0.931	0.855 (text-calibrated)

575 **The in-distribution shortcut, quantified.** The numbers behind §4.1’s claim that the final-layer
576 probe’s in-distribution AUC of 0.822 is mostly pool identity: a linear read of the same hidden state
577 classifies a query’s source pool at 95.8% accuracy; a pool-identity oracle alone reaches AUC 0.715;
578 and appending true pool labels to the probe’s features moves it 0.822→0.821. In-distribution area
579 over random is statistically on par between the gate and the pool oracle (Appendix E); the two
580 separate only under LOPO, where the probe inverts to 0.372 and the oracle is undefined.

581 **ROC curves.** Figure 3 shows the calibration-split ROC for the signals of Table 3.

582 $p(\text{True})$ **backbone dependence.** Across backbones, post-draft $p(\text{True})$ beats the trained final-
583 layer probe on four of five raw/duplex models tested; the exception (GLM4-9B-chat, 0.685 vs. probe
584 0.798) shows the pattern depends on the backbone’s self-evaluation quality.

585 **Matched pairs.** Table 4: with label coverage matched by subsampling the raw control to the
586 duplex model’s per-pool fail rates, the raw backbone keeps LOPO math at 0.825/0.809 while duplex
587 models sit at chance — label coverage is ruled out; the representation changed. Degradation is
588 ordered raw > omni-streaming > duplex. Table 5 gives the by-depth summary behind Figure 2:
589 the within-pair deficit at the best mid-layer is only −0.03 vs. −0.59 at the final layer; mean-pooled
590 probes survive to the final layer on duplex models (LOPO math ≈0.80 at L35 on o4.5, 0.674 at L27
591 on o2.6 — 0.13–0.15 below each model’s mid-layer last-token peak, and uniformly weaker in-mix,
592 −0.058 at L22); duplex damage is severe but local, omni-streaming damage mild but diffuse.

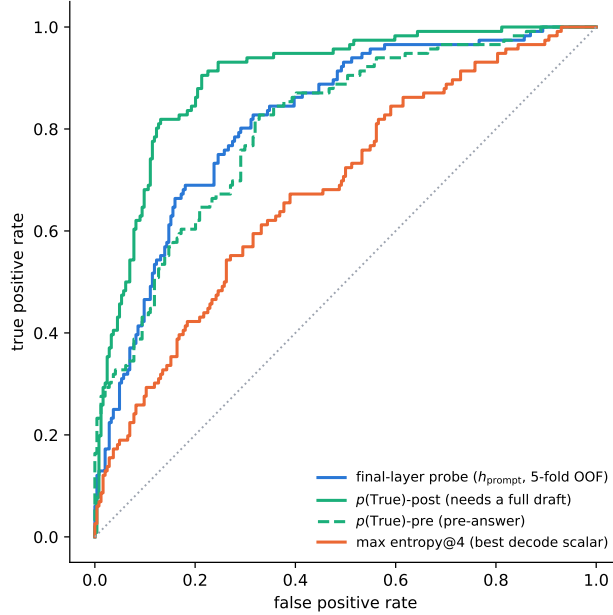


Figure 3: ROC on the calibration split. $p(\text{True})$ -post leads but needs a full drafted answer; the probes and $p(\text{True})$ -pre are pre-answer.

Table 4: LOPO transfer (AUC, held-out pool) of the final-layer last-token probe, Qwen2.5-7B family, label-coverage matched.

LOPO pool	Qwen2.5-7B (raw)	Qwen2.5-Omni	MiniCPM-o 2.6	[o4.5 vs Qwen3]
hard-math	0.809	0.745	0.538 $\approx$ chance	0.372 (inverts)
hard-knowledge	0.680	0.613	0.526 $\approx$ chance	0.61

Modality specificity of the cliff. Under audio input the final layer does *not* invert (L35 audio LOPO math 0.936 vs. 0.366 text); a speak-mode template control on text input leaves the cliff unchanged (0.362) — the operative variable is the modality of the input context, not the output mode.

Audio collapse of verbalized introspection. Table 6. Three controlled arms pin the mechanism on MiniCPM-o 4.5: *not perception* (ASR audit WER 0.074; $p(\text{True})$ on the model’s own transcript snaps back to 0.074; the well-heard subset still collapses); *not a context prior* (irrelevant filler audio does not inflate p_{yes}); *binding* (duplicating the question as text alongside the audio fully restores introspection, 0.034). The verbal instance-check runs over text-token pathways that audio embeddings do not feed, while a linear probe reads the same information off the same forward pass at 0.93+. The failure implies its own cheap fix, confirmed twice: re-present the query as text before asking (“repeat-then-judge”).

Probe $\oplus$ repeat-then-judge fusion. The restored self-evaluation is *additive* with the probe. We stack the deployed L22 probe with $p(\text{True})$ -pre asked over the model’s own transcript (one short text prefill + a 1-token decode per turn; logistic stacker fit on the 360 calibration rows, native regime, Table 7). On the internal test split the fusion is the first significant AUC lift over the deployed probe (0.836 $\rightarrow$ 0.862; bootstrap 95% CI on Δ excludes zero), and it transfers to one external pool (TriviaQA 0.789 $\rightarrow$ 0.820); the remaining English pools move positively but within CI, and the Mandarin reasoning pool is untouched — there the self-evaluation itself carries no signal (AUC 0.586). Contrary to our a-priori expectation, the gain tracks where the self-evaluation signal is strong, not where

Table 5: LOPO math transfer by depth (last-token probes).

model	best mid-layer	final layer	shape
Qwen3-8B (raw)	0.964 (L33/36)	0.958	plateau, no cliff
MiniCPM-o 4.5 (duplex)	0.931 (L22/36)	0.366	cliff, inverts
Qwen2.5-7B (raw)	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni (streaming)	0.794 (L16/28)	0.746	diffuse depression
MiniCPM-o 2.6 (duplex)	0.822 (L21/28)	0.540	cliff

Table 6: Pre-answer introspection under audio input. Trap p_{yes} = mean stated probability of answering correctly on a 100%-fail pool (lower is better).

model	FT type	trap p_{yes} : text $\rightarrow$ audio	audio $p(\text{True})$ -pre AUC
MiniCPM-o 4.5	duplex	0.055 $\rightarrow$ 0.556 (collapse)	0.786 (trap dead)
MiniCPM-o 2.6	duplex	0.196 $\rightarrow$ 0.345	0.491 $\approx$ chance
Qwen2.5-Omni	omni-streaming	0.279 $\rightarrow$ 0.213 (intact)	0.727
Freeze-Omni	frozen + adapter	0.165 $\rightarrow$ 0.112 (intact)	0.612 (capability caveat)

the probe is weak; a diagnostic arm asking $p(\text{True})$ -pre on the original query text scores within 0.01–0.02 of the transcript arm on every pool, so transcription quality is not the limiter.

Table 7: Probe $\oplus$ repeat-then-judge $p(\text{True})$ -pre fusion (native regime, deployed read point). Δ = fusion – probe, bootstrap 95% CI.

pool	n	probe	rtj solo	fusion	Δ [95% CI]
internal test	240	0.836	0.836	0.862	+0.026 [+0.004, +0.048]
TriviaQA	250	0.789	0.761	0.820	+0.030 [+0.000, +0.062]
WebQ	250	0.745	0.723	0.771	+0.026 [−0.007, +0.059]
Llama-Q	250	0.686	0.674	0.691	+0.006 [−0.020, +0.031]
SD-QA	200	0.803	0.742	0.808	+0.004 [−0.023, +0.032]
Reasoning-zh	202	0.640	0.586	0.618	−0.021 [−0.076, +0.032]

Real-speech validation. The matched-pair design rerun on 200 real human recordings (VoiceBench SD-QA, USA split) replicates all three audio-side findings: modality tax (0.400 $\rightarrow$ 0.450), audio overconfidence (paired p_{yes} shift +0.089; failure subset 0.415 $\rightarrow$ 0.581), and the layer structure (text-calibrated read on real-speech audio holds 0.76–0.80 at L22–L26, peak 0.800 at L25 — transfer across both content and modality).

Synthesis. End-to-end multimodal training *builds* the shared mid-layer core; duplex-style training additionally *damages* the readouts; omni-streaming gets both right; an adapter alone gets neither (probe readout text-fragile, verbal readout audio-fragile; in both cases knowledge survives and a readout breaks, and in both cases the breakage is duplex-fine-tune-specific).

The deployed probe, and which calibration each result uses. The mechanistic probe is a single-position L22 read (4096-d), fit on the 360-query calibration split; every representational result (§4, Appendices B, D) is this probe. The deployed probe adds two zero-latency pooled positions (an 8-token tail mean and the running user-audio mean; 12,288-d), refit on a calibration mix widened to 2,310 public-benchmark queries disjoint from every evaluation benchmark; a pre-registered guard held (frozen internal test AUC 0.860 $\rightarrow$ 0.879) and feature selection never saw the externals. It drives the turn-based live loop (§6), the frozen transfer results (§7), and the talker-on replay — frozen throughout, calibrated on text and speech-rendered internal data only. The serving-regime validations are the exceptions. In the concurrent regime, the same 12,288-d feature set is refit in-regime on 360 rows for the loop of Table 14 and on the full 2,310-row mix to measure the effect of calibration scale (Appendix H). In the native regime, the same feature set is refit on 2,310 rows collected at the native listen $\rightarrow$ speak read point (Appendix O). *Selection history.* L22 and the three-position feature

set were chosen on internal calibration-split cross-validation before any external benchmark was scored (the 8-position $\times$ layer sweep and the 19-configuration feature sweep are both calibration-only), and the internal frozen test split was the pre-registered guard. The external pools were then read once with the frozen artifact. Two honest exposures remain: the layer band was picked on internal LOPO rather than by nested selection inside each held-out pool, and the second duplex family’s mid-band peak (L30–34 of 56) was located with its own sweep, so both layer choices are best read as *confirmed on held-out pools*, not as untouched-holdout estimates. In the mid-layer band the in-mix tradeoff optimum is broad (L20–L30) and the mid-layer probe’s in-distribution area advantage is modest — its real advantage is LOPO robustness, which an in-mix test cannot show.

C Second duplex family: NemotronLabs-VoiceChat-11B

The pre-registered prediction test: with the identical recipe (same audio, same judge, 600-query calibration), the layer sweep on this hybrid-Mamba duplex model (56 blocks: 27 Mamba-2 / 4 attention) peaks mid-band at L30–34 (eot-last AUC 0.714 vs. 0.693/0.682 at the endpoints); stacking the same three positions lifts OOF 0.714 $\rightarrow$ 0.761 $\rightarrow$ 0.790; the frozen-methodology probe transfers to the external pools at AUC 0.781/0.793/0.754/0.701 — the deployed-probe band reached with a quarter of the calibration data. Boundary notes: the backbone is much weaker at knowledge retrieval (striviaqa local 0.32 vs. MiniCPM 0.62, same judge) and English-only (the cross-lingual pool has no analog). The live-loop port and the trained-checkpoint ablation remain future work; the port is gated by the released inference path, not by the architecture. The only supported loading path (the NeMo Speech nemotron-labs-voicechat branch) runs the duplex loop *cacheless* — every 80 ms frame re-runs the full prefix, an $O(T^2)$ schedule that cannot hold the frame clock in real time — although the hybrid backbone’s Mamba-2 state is exactly the $O(1)$ per-frame recurrence a streaming port would need; wiring stateful inference into the released duplex frame protocol is the engineering gap. (Offline, the same property is a convenience: the final frame’s forward contains every position, so one hook captures the whole eot window.) Accuracy-side conclusions do not wait on the port: the re-mix arithmetic below tracks measured live arms at 0.011 mean deviation on MiniCPM, inside the ± 0.02 –0.03 replication floor.

Escalation re-mix on every transfer pool. Completing the grid of Table 1 on this family: an offline re-mix (the same top- r reconstruction arithmetic that, on MiniCPM, tracks the measured live arms at 0.011 mean deviation) sends the top- r of each pool by NVDA probe score to the measured GPT-5.5 outcome and keeps NVDA’s own answer for the rest, scored with each pool’s official protocol (the OAB judge for the three OpenAudioBench pools — NVDA local floors 0.352/0.376/0.705 — our judge for SD-QA, VoiceBench 1–5 for AlpacaEval; the expert answers the heard transcript and is judged directly, so the relay-back channel is excluded — including the measured relay outcomes where they exist shifts tier points by ≤ 0.032). **Selective escalation beats matched-rate random on all five pools** (permutation $p \leq 0.0004$ at the 30% and 50% budgets, $p < 0.03$ at 15%; Table 8, Figure 4) — including Web Questions, SD-QA and AlpacaEval, the three pools that are flat or negative on MiniCPM (§7). Selectivity is nearly unchanged (AUC 0.72–0.78 under official labels); what changed is headroom — floors of 0.31–0.38 against MiniCPM’s 0.51–0.66 — and on AlpacaEval the failure species: the terse English-only model fails open-ended queries *hard* (the probe-selected half scores 3.01 vs. 4.08 kept), where MiniCPM’s open-ended failures are soft quality gradations a wrongness detector cannot see. The negatives are properties of the pool $\times$ backbone pairing, not of the signal.

D What breaks at the output: mechanism audit

§4.2 reports *that* the final-layer last-token read inverts. This section reports what we could and could not establish about *why*. Everything below uses the Phase-5d all-layer prefill dumps (every decoder layer, last-token and mean-pooled, same 600 queries) and the identical probe protocol as the layer sweep: calibration split, L_2 logistic regression, LOPO trained on the four non-math pools and scored

Table 8: The frozen-recipe NVDA probe re-mixed at fixed escalation rates, official judge per pool, each against its own matched-rate random reference (same measured outcomes, escalation set drawn at random; identical at 0% and 100% by construction). AlpacaEval rows are VoiceBench 1–5 scores; others accuracy. Every pool clears random at all three budgets: $p \leq 0.0004$ at 30% and 50%, $p < 0.03$ at 15% (20k permutations, `scripts/18_nvda_random_check.py`).

pool	judge	n	0%	15%	30%	50%	100%
Speech TriviaQA matched random	OAB	247	0.356	0.498 0.446	0.615 0.536	0.741 0.656	0.955
Speech Web Questions matched random	OAB	250	0.376	0.480 0.444	0.588 0.512	0.696 0.602	0.828
Llama Questions matched random	OAB	234	0.705	0.786 0.739	0.838 0.773	0.876 0.818	0.932
SD-QA matched random	ours	200	0.310	0.425 0.397	0.545 0.484	0.675 0.600	0.890
AlpacaEval (1–5) matched random	VB	199	3.55	3.83 3.75	4.13 3.96	4.46 4.23	4.92

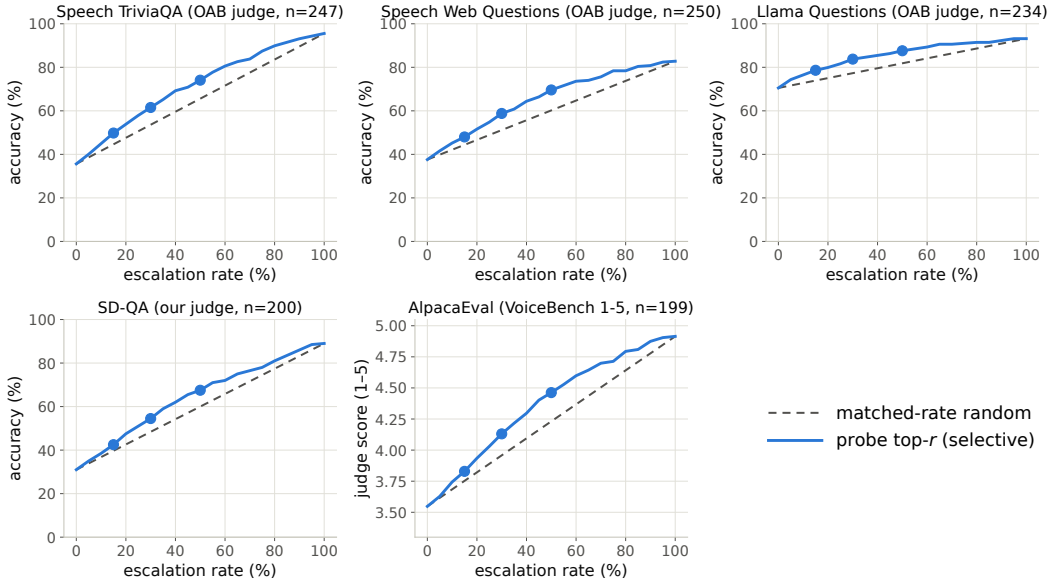


Figure 4: Re-mix curves for the frozen-recipe NVDA probe: selective (top- r by probe score) vs. matched-rate random on the five transfer pools, official judge per pool. Every pool separates, including the three that are flat on MiniCPM (all five pools, $p \leq 0.0004$ at 30/50% and $p < 0.03$ at 15%, permutation).

684 on held-out math. The reproduction check recovers the published L22 and final-layer numbers to
685 ≤ 0.012 (`scripts/14-17_*.py`).

686 **Four mechanisms ruled out.** Table 9 lists them. (i) *Speak mode*. Prefilling the same text queries
687 under the TTS/speak-mode template leaves the cliff exactly where it was (final-layer LOPO math
688 0.362 vs. 0.366 plain), so the readout is not being displaced by a prepare-to-speak state. (ii) *Modality*
689 *axis*. The mean audio-minus-text direction at the final layer is orthogonal to both the inverting final-
690 layer probe and the transferring L22 probe (cosine 0.013 and 0.027; chance is $1/\sqrt{4096} = 0.016$), so
691 the cliff is not the audio/text split that §4.3 measures. (iii) *More shortcut available late*. Source-pool
692 identity is linearly decodable at 0.88–0.96 at *every* depth, in the duplex model and its raw backbone
693 alike — the query-type shortcut of §4.1 is not something the late layers add; it is what remains once
694 the competence signal stops being readable. (iv) *Wholesale representational drift*. Raw linear CKA

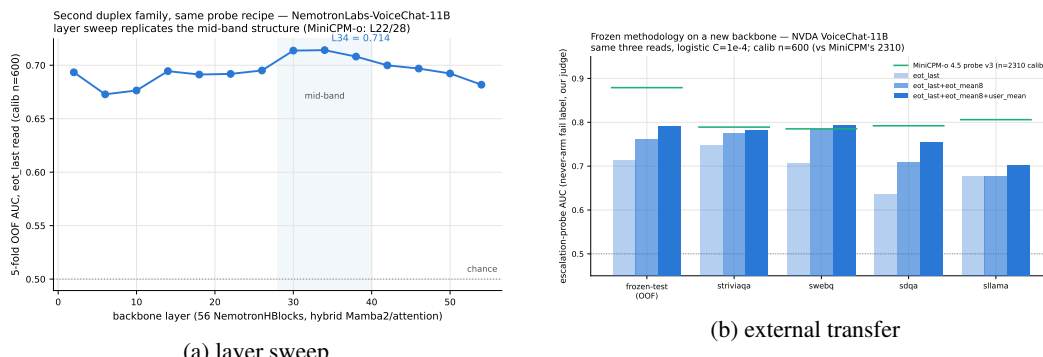


Figure 5: Second-family test: (a) the usable read again peaks mid-network; (b) frozen-recipe transfer lands in the deployed-probe AUC band.

695 between the duplex model and its backbone appears to collapse late (0.80→0.47), but transformer
 696 residual streams carry a few massive-activation coordinates that dominate the Gram matrix; with
 697 per-feature standardization the same comparison is nearly flat (0.82→0.78) and does not localize
 698 the cliff. Projecting out the massive-activation axes, or the pool-mean directions, rescues nothing
 699 (0.35–0.40 at every $k \leq 20$, against a random-direction control of 0.37).

Table 9: Mechanism candidates for the final-layer inversion. Final-layer LOPO hard-math AUC on MiniCPM-o 4.5 unless noted; 0.366 is the unmodified baseline and 0.931 the L22 read.

candidate	test	verdict
prepare-to-speak state	speech-mode template prefill	0.362, no change
modality (audio vs. text) axis	cosine with the probe direction	0.013 $\approx$ chance
late layers encode more pool identity	pool decoding by depth	0.88–0.96 at all depths
representation rewritten late	standardized CKA vs. backbone	0.82→0.78, flat
massive-activation coordinates	project out top- k such axes	0.35–0.40, no rescue
query-type directions	project out pool-mean directions	0.34–0.45, no rescue
delivery to the output position	residual component of the read	0.93→0.27 (Fig. 6)

700 **What does separate the models.** Figure 6 measures the probe rather than operating on the model.
 701 For the probe trained at each depth on the four non-math pools, we split its held-out score into the
 702 part carried by that layer’s five dominant principal directions (estimated from training rows only)
 703 and the residual, $s = w^\top Px + w^\top (I - P)x$. The residual component is the distributed read, and
 704 it is the one that transfers: it holds 0.85–0.93 through mid-network on MiniCPM-o 4.5 and then
 705 decays to **0.27** at the output, with the same decay on MiniCPM-o 2.6 (0.81→0.52) and *no* decay
 706 on either raw backbone (Qwen3-8B 0.86, Qwen2.5-7B 0.71 at their final layers) or on the omni-
 707 streaming control (Qwen2.5-Omni holds 0.75 to its output with a flat variance share) — the decay
 708 tracks duplex fine-tuning specifically, not streaming or audio fine-tuning in general. On MiniCPM-
 709 o 4.5 the score also concentrates: the dominant subspace carries 39% of the held-out score variance
 710 at the final layer against ≤ 0.18 at every earlier depth and 0.05 for its backbone; on MiniCPM-o 2.6
 711 this concentration is present but weak (0.16 vs. 0.14 for its backbone), so we report it as a property
 712 of the stronger duplex model, not a general law.

713 **Direct turn-control tests.** Two experiments target the turn-control story head-on rather than by ex-
 714 clusion (modal_interp.py, scripts/19_*.py). *Control-token logit lens*: applying each model’s
 715 own final norm and unembedding to the stored per-layer states, the probability mass on the trained-in
 716 control vocabulary (added/special tokens) *rises* over MiniCPM-o 4.5’s last layers (log-mass −23.8
 717 at L32 to −11.8 at the output) while the raw backbone’s falls by fifteen log-units over the same span
 718 — the duplex output layer, uniquely, stays ready to emit control tokens. But the fine-grained predic-
 719 tion fails: the inverting probe direction is no more aligned with the control tokens’ unembedding

rows than with random vocabulary rows ($\max |\cos|$ 0.069 vs. 0.068), and per-query control mass correlates with the probe score at only $r=0.18$. *Listen/speak intervention*: re-prefilling 60 calibration queries with a still-listening vs. answer-now suffix moves the final-layer state substantially in *both* models (relative displacement 0.27 duplex, 0.59 raw — cue text is just prompt semantics), and the cue direction is orthogonal to the inverting probe ($|\cos|=0.019$, chance 0.016), shifting its score by only $+0.23$ sd (and the deployed L22 read by $+0.08$ sd). Together: the duplex output layer is measurably control-biased in its output distribution, but the inversion is not carried by control-token directions or by an explicit listen/speak axis — the simple takeover story fails its own direct tests.

What we do not claim. Destructive tests do not settle the mechanism. Removing the top five principal directions from the final layer does raise LOPO math from 0.378 to 0.758 — but the identical operation costs the raw backbone $0.90 \rightarrow 0.14$ and costs the intact L22 read $0.931 \rightarrow 0.760$, so it is a damaging intervention that happens to help a broken readout, not evidence of a duplex-added subspace. We therefore state the finding at the level the evidence supports: duplex fine-tuning does not erase the competence signal, it stops delivering it in distributed form to the final last-token position, and the layer where delivery fails is also the one whose output distribution stays control-biased. Attributing the loss to a specific duplex objective needs training-time access we do not have; the practical consequence — read mid-network — does not depend on the attribution.

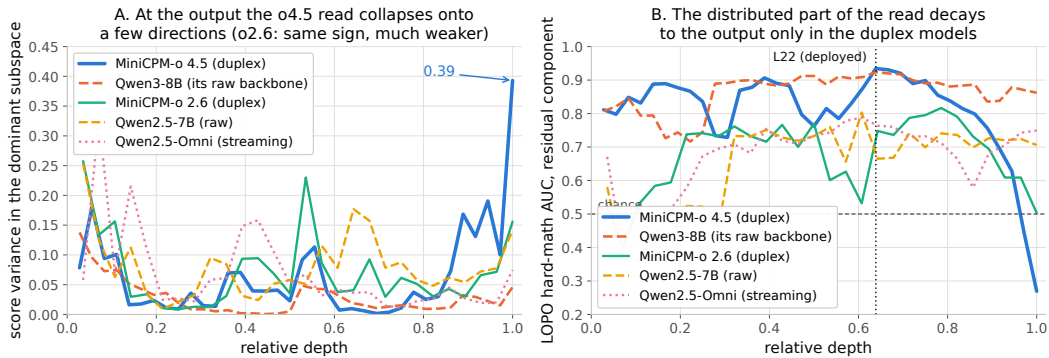


Figure 6: Non-destructive decomposition of the probe’s held-out score at every depth. **Left:** share of score variance carried by the layer’s five dominant directions. **Right:** transfer AUC of the residual (distributed) component — intact to the output on both raw backbones and on the omni-streaming control, decaying to inversion over the final layers of both duplex models.

737 E Query-surface router audit

A RouteLLM-style TF-IDF $\rightarrow$ LR router under identical labels reaches OOF AUC 0.669 (below the pool oracle’s 0.715); a 24-configuration sweep tops out at 0.689, so this is the query surface’s ceiling, not tuning. Under LOPO it collapses to 0.38–0.57 and scores the 100%-fail trap pool at 0.230 — it misses the pool a router most needs. The training receipt: at the argmax threshold its out-of-fold accuracy exactly matches the majority class; the internal probes under the same accounting clear majority by 9–29 points (Figure 8). Nor is it data-starved: the identical recipe on RouterBench [Hu et al., 2024] (36,497 prompts) reaches only in-domain AUC 0.710 with leave-one-benchmark-out at chance; RouteLLM’s released preference-trained routers score 0.523/0.533 zero-shot on our pool and rank the trap pool below ordinary hard pools; our router transfers to RouterBench below chance (0.440). On audio labels the router improves (test 0.814) but still trails the mid-layer probe (0.879) and cannot fire mid-stream. In distribution, the gate’s area over random ($+0.054$) is statistically on par with the pool oracle’s ($+0.042$; paired bootstrap delta CI $[-0.003, +0.027]$, $p=0.13$) — the gate’s case is transfer, not the in-distribution margin. $p(\text{True})$ tradeoff variants: Figure 9.

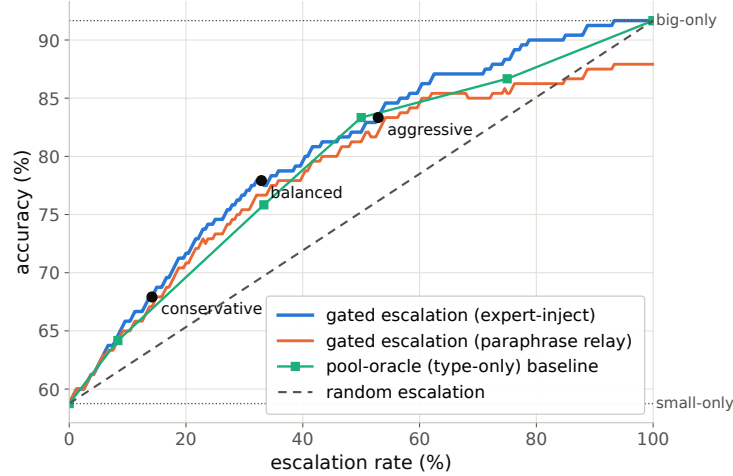


Figure 7: Accuracy vs. escalation rate, frozen test split: gated escalation vs. random-escalation and pool-oracle (type-only) baselines between the small-only and big-only endpoints.

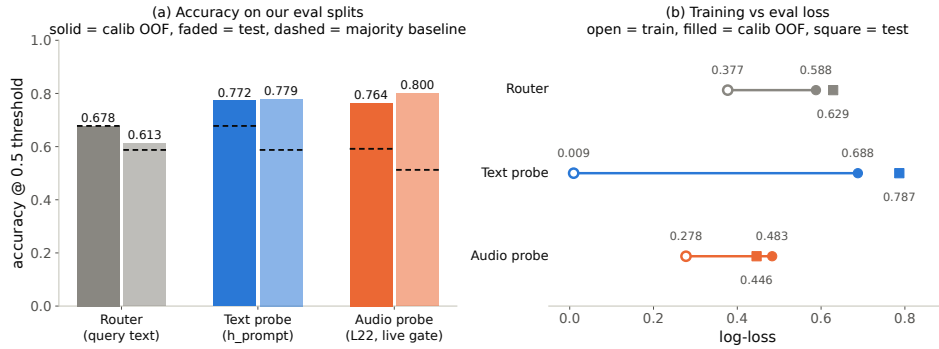


Figure 8: Identical training receipt for the external router and the two internal probes: the router never clears the majority-class baseline; both probes do, by 9–29 points.

F Gate timing details

Against the model’s own thinking mode. MiniCPM-o 4.5 ships a built-in reasoning mode — self-escalation in place. Gated-cloud escalation dominates all-THINK on both accuracy and latency at every escalation rate (gated-cloud@33%: 78.7% at mean 6.5 s vs. all-THINK 63.7% at 22.4 s; Table 10). The reason is mechanistic: the gate flags knowledge and trap failures, which extra compute cannot fix — so external escalation is necessary, not merely better.

Table 10: Deployment policies, frozen test split ($n=240$); latencies measured.

policy	acc (%)	lat. mean (s)	P50	P95
fast-only	58.8	3.0	3.5	4.2
all-THINK (built-in)	63.7	22.4	17.2	60.1
gated-think @33%	61.3	12.1	3.6	47.6
gated-cloud @15%	68.8	5.3	3.6	10.1
gated-cloud @33%	78.7	6.5	3.6	20.8
gated-cloud @50%	85.8	7.0	3.6	24.4

Measurement protocol. Gate latency is wall-clock from prefill start to the L22 score being available (truncated forward + probe dot product), CUDA-synchronized, one H100, 50 queries per

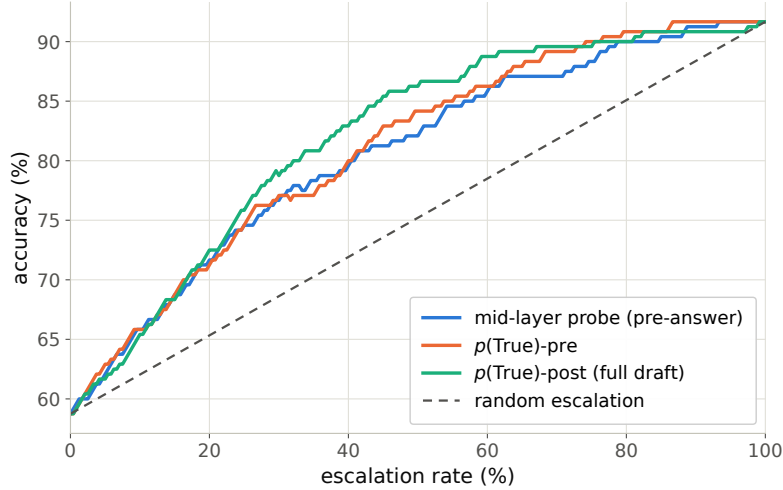


Figure 9: Offline tradeoff for the $p(\text{True})$ signals (pre-answer and post-draft).

modality, three warmup calls excluded, interleaved per query with the reference arms; P50/P95: gate 20/25 ms text and 45/104 ms audio vs. full-prefill TTFT 36/47 and 68/144 ms. Network and expert-RPC initiation are excluded: the cloud call fires asynchronously at the gate, and its user-facing cost is reported as expert-content-audible latency in Appendix G. Absolute prefill wall-times vary with call overhead across benches; the robust cross-bench statistic is the ratio — the L22 state is ready at $\sim 57\text{--}61\%$ of prefill.

Per-layer truncated-forward cost vs. quality (Figure 10): the L22 hidden state is available at $\sim 57\text{--}61\%$ of prefill wall-time while L22 is simultaneously the in-mix quality peak; forking much earlier is both weaker and modality-specific, so the fork belongs at $\sim 55\text{--}60\%$ depth. If the cloud call launches at the gate, the expert result arrives before the local answer finishes for 40–58% of the whole mix but only $\sim 20\text{--}31\%$ of *escalated* queries (they skew to short local answers); the residual wait is P50 2–3 s, P90 ~ 27 s (Figure 11) — the stall-phrase budget the injection protocol must cover.

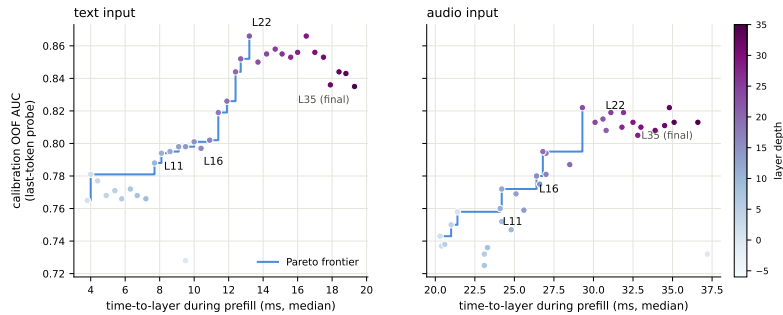


Figure 10: Fork-depth Pareto: per-layer wall-time vs. gate quality.

G Live-loop details

What the signal detects. The pre-answer probe is specifically a *retrieval-failure* detector, strongest on missing-knowledge and long-tail-trap failures where the empty lookup is observable in the model’s state. Execution failures (stuck mid-derivation) surface in *decode-time* signals instead, and metacognitive failures — false-premise questions, where no failure event occurs at all — are invisible to every signal we tested, query-surface routers included (fire rate 18% vs. trap’s 90%; Appendix G), which motivates a decode-time check behind the pre-answer gate.

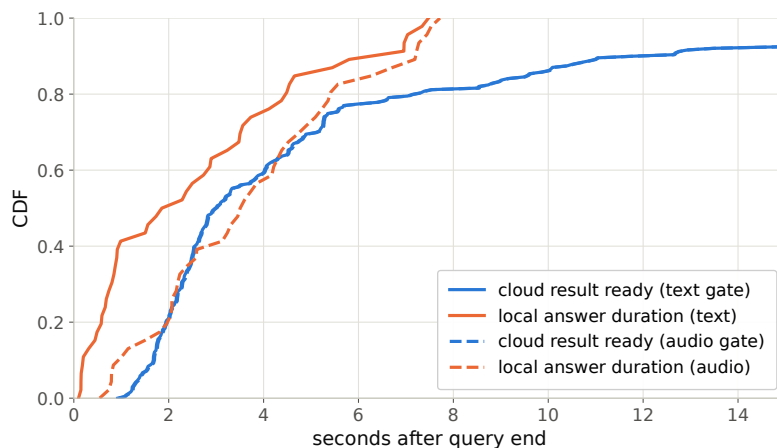


Figure 11: Cloud round-trip vs. local answer duration: probability the expert result arrives before the local answer ends.

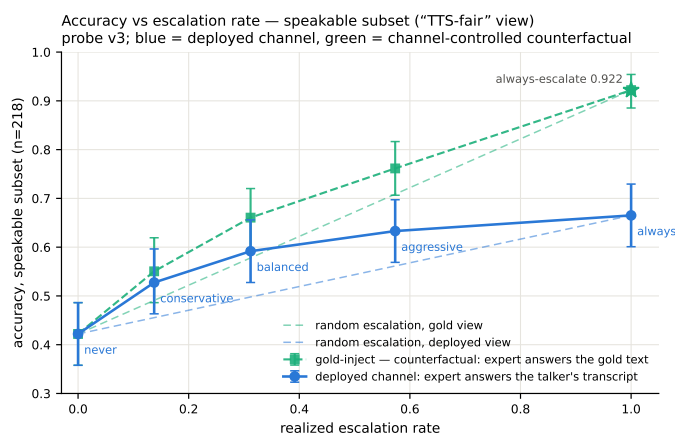


Figure 12: Live dual-view tradeoff ($n=218$ speakable subset). Blue: measured deployed-channel accuracy of the live system (paired-bootstrap CIs). Green: the channel-controlled counterfactual — same sessions, escalated rows re-scored with gold-text expert answers. Each view is plotted against *its own* matched-rate random reference (faint dashed, same floor, each view’s own 100% endpoint): selection beats random within both views at every arm, by 0.072–0.095 deployed and 0.054–0.084 gold. The vertical view-to-view gap is the speech-channel price; the floor-to-floor gap to offline operation is the audio + streaming-answer tax.

778 **Full live tables.** Table 11 (both scoring views, speakable subset and full pool) and Table 12 (measured response latency). The conservative tier *shortens* the P95 tail: its few escalations displace exactly the longest local decodes. Figure 13 joins accuracy and latency; Figure 14 replays three real sessions from recorded timers.

782 **Time to first audible audio.** The sweep above is text-out; a TTS-on re-run of the balanced arm (same 240 queries, talker head + vocoder on, stall pre-synthesized once in the talker’s own voice) measures speech onset directly. The gate decision is unchanged — the TTS token enters the context only after the end-of-turn read (read p50 21 ms / p99 32 ms), and the same 84/240 queries fire. From gate decision to first audible audio: **local answers p50 0.552 s / p95 0.599 / p99 0.644; escalated turns p50 0.022 s / p99 0.029** — the canned stall (3.2–4.1 s of speech) plays the moment the gate fires, so the escalated path reaches first audio 25× faster. Expert *content* becomes audible at p50 6.4 s / p95 26.0 / p99 34.0 (cache-corrected expert round-trip + first relay PCM, p50 0.66 s after the

answer returns); per-turn dead air between stall end and expert content is p50 2.5 s / p95 22.4 / p99 30.7, with 25/84 escalated turns fully covered — the stall covers the head of that wait, and Table 12’s completion profile bounds the tail.

Table 11: Live streaming sweep, paired bootstrap 95% CIs ($B=10^4$). Top: speakable subset ($n=218$); bottom: full pool ($n=240$). Gold-inject = escalated rows re-scored with gold-text expert answers; channel cost = paired difference. * CI excludes 0; † CI lower bound at zero.

tier	esc	deployed	Δ vs. floor	gold-inject	channel cost
never	0%	0.422 [0.36,0.49]	—	0.422	—
conservative	14%	0.528 [0.46,0.59]	+0.106*	0.550	+0.023†
balanced	31%	0.592 [0.53,0.66]	+0.170*	0.661	+0.069*
aggressive	57%	0.633 [0.57,0.70]	+0.211*	0.761	+0.128*
always (measured)	100%	0.665 [0.60,0.73]	+0.243	0.922 [0.89,0.95]	+0.257
never (full)	0%	0.383 [0.32,0.45]	—	0.383	—
conservative	16%	0.483 [0.42,0.55]	+0.100*	0.525	+0.042*
balanced	35%	0.554 [0.49,0.62]	+0.171*	0.654	+0.100*
aggressive	61%	0.596 [0.53,0.66]	+0.212*	0.771	+0.175*
always (measured)	100%	0.617	+0.234	0.917 [0.88,0.95]	+0.300

Table 12: Measured response latency (s), $n=240$ per arm (latency is channel-independent). P99 on $n=240$ is indicative only.

tier	mean	P50	P95	P99	Δ mean	Δ P50	Δ P95	Δ P99
never	4.6	2.0	15.8	17.8	—	—	—	—
conservative	4.6	2.6	12.8	23.0	+0.0	+0.6	−3.0	+5.2
balanced	5.8	3.5	15.9	32.7	+1.2	+1.5	+0.1	+14.9
aggressive	6.6	4.4	18.5	33.0	+2.0	+2.4	+2.7	+15.2

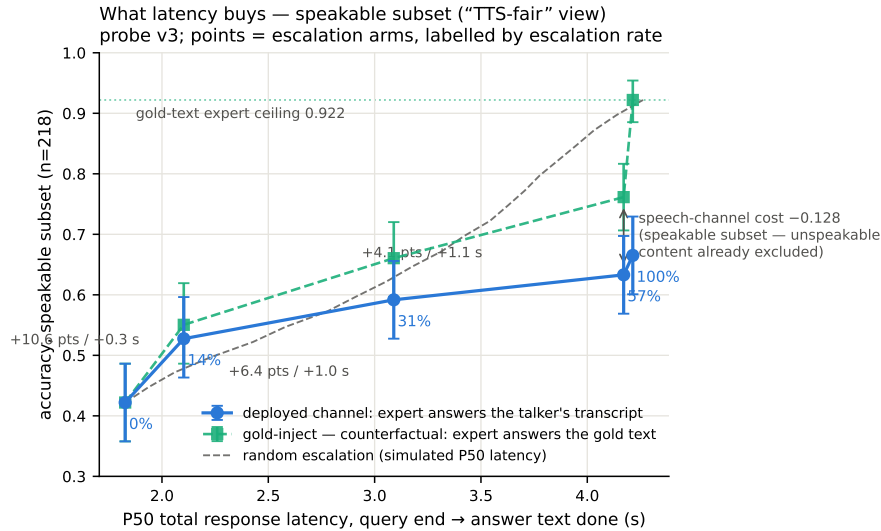


Figure 13: Accuracy vs. per-arm P50 latency (speakable subset), both scoring views, with the simulated-P50 random reference.

Speculative prefetch, measured and dropped. Framed as speculative escalation (draft–verify at sentence granularity), its acceptance rate — does an expert answer to the partial transcript answer the full question — is 0.07/0.19/0.51 at 25/50/75% of the audio; at the mid-stream gate’s typical fire point 6–8 of 10 calls would be wasted. The trap pool’s floor (0.00/0.10/0.27) shares the back-loaded-core property that breaks mid-stream probe reads.

Three live sessions, timestamped (balanced arm; every event time is a recorded live-loop timer)

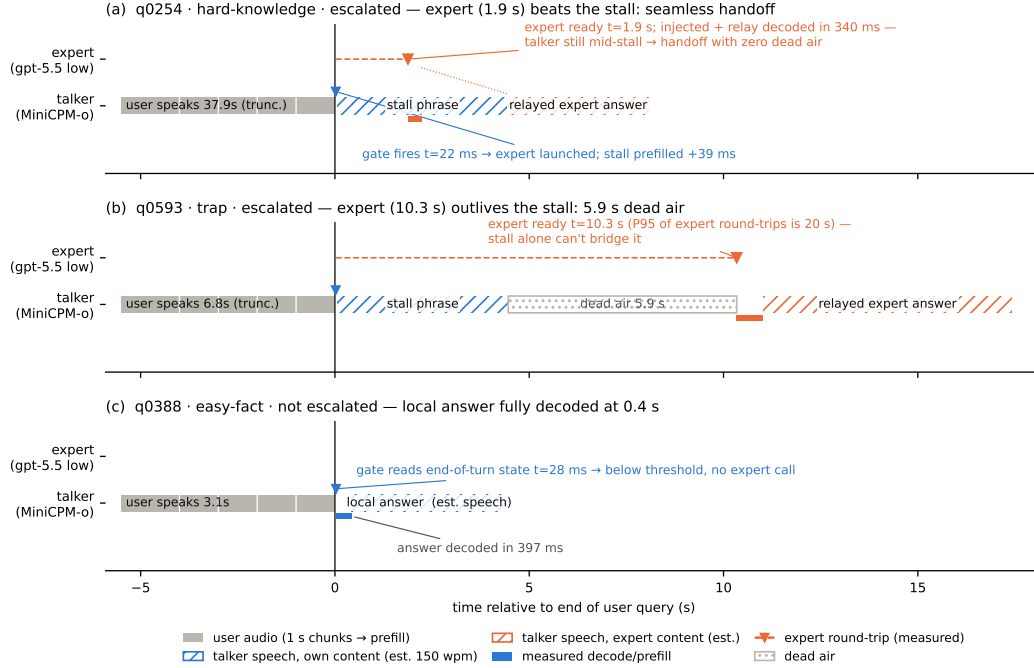


Figure 14: Three real sessions from recorded timers: expert beats the stall phrase (seamless); expert outlives it (dead air, the P99 signature); gate stays closed (local answer fully decoded 0.4 s after end of turn). Speech occupancy is estimated at 150 wpm (hatched); the measured arms are text-out, so no vocoder onset is plotted.

798 **Conflicting-injection controls (172 sessions).** Fabricated conflicting expert answers by within-
799 pool derangement, four framings: worst-case comply is 0.00 neutral / 0.53 deployed authority / 0.79
800 strong-override, while *seeding* words into the talker’s mouth backfires (comply 0.25, lip-service 0.62
801 — forced speech is treated as revisable). Math is silently re-derived rather than relayed or disputed.
802 Model-wrong × wrong-injection swallow rate is 0.64: expert quality is the accuracy ceiling of the
803 cascade.

804 **Uplink diagnostics.** On the escalated population the expert answering the talker’s own transcript
805 scores 0.585 vs. 0.871 on gold text. Prompt-level rescues on the self-transcript are no-ops (the
806 damaging errors are internally-coherent entity substitutions); replacing the ears helps: open-weights
807 Whisper recovers little (+4 pp) but a stronger cloud ASR on the same audio recovers 38% of the gap
808 (0.585→0.694, McNemar $p=0.007$) — diagnostic only (its critical-path latency and external-pool
809 survival are unmeasured). A same-model gold-text control with an audio-capable expert shows the
810 audio channel itself is lossless for speakable content: the leak is the talker’s transcription, not the
811 synthesis or the relay (relay drops: 5 cases of 108; corrupted transcripts dominate the escalated
812 failures, first-generation sweep).

813 **Boundary query classes.** *False premises:* the model fails substantially (0.63 text / 0.47 audio)
814 and the expert would help (0.80), but the gate is blind (fire@balanced 0.18 vs. trap’s 0.90; no score
815 separation) — answering along a false premise does not feel like inability. *Real-time queries* (“what
816 is NVDA trading at”): the gate is not blind, the class was simply absent from every static training
817 family; extending calibration with FreshQA [Vu et al., 2023] (fast-changing labeled escalate a priori,
818 never-changing as in-family controls) raises held-out fast-changing fire rates to 0.80/0.93/1.0 across
819 tiers with frozen-test AUC unchanged (0.877 vs. 0.879). One calibration subtlety: tier budgets must

820 remain quantiles of the deployment-mix scores, or the new capability is spent on its own threshold
821 shift.

822 **Demonstration system.** The pipeline runs as an interactive web demo on one H100, on the
823 model’s *native full-duplex* head (Appendix O): every second of microphone audio is prefilled into
824 the context the talker is generating from, the head itself decides listen/speak per chunk, and the
825 gate reads L22 at the listen→speak transition against the in-regime thresholds. Escalated turns play
826 a canned stall (teacher-forced once, the talker’s own voice), consult the expert (with a demo-only
827 web-search tool for real-time queries), and the verified answer re-enters the stream as a text unit
828 that the talker voices — an ordinary duplex turn, hence interruptible like any other. There is no
829 VAD, no ASR barge-in classifier, and no abort machinery: floor control is entirely the duplex head’s
830 own behavior (Appendix O). An earlier demo build approximated barge-in with an energy-VAD +
831 burst-ASR harness over the turn-based loop; it is retired, and survives only as the harness-overlap
832 arm measured above. The demo will be made available upon publication.

833 H Duplex-validation sweep: the gate under the talker

834 The headless live sweep (§6) never engages the talker head; this sweep replays its full 240-query
835 pool through the deployed voice loop above to test whether the frozen probe read survives the spo-
836 ken regime. Two arms, one session per query, deployed v4 gate with scores logged but escalation
837 disabled (the arm isolates gate validity, not routing outcomes). **Clean:** the query streams as mic-
838 shaped PCM (pool TTS audio plus steady noise at RMS 0.008), the energy VAD ends the turn, the
839 talker answers aloud, and the session is cut after the first audible chunk. **Overlap:** an easy warmup
840 question (probe score <0.25 , locally answered, >5 s spoken answer) puts the talker mid-answer,
841 and the target query is then spoken over it; sustained overlapping speech commits a barge-in whose
842 audio seeds the next turn. The barge-in path engaged on 237/239 pairs.

843 Against the frozen local-failure labels of the same pool, the identical frozen probe reads: headless
844 AUC 0.866 [0.817, 0.909], clean 0.871 [0.824, 0.914], overlap 0.856 [0.803, 0.901]; barged-only
845 subset ($n=237$) 0.854. Paired bootstrap deltas vs. headless are $+0.005$ [$-0.025, +0.035$] (clean) and
846 -0.009 [$-0.036, +0.016$] (overlap) — both inside the ± 0.02 – 0.03 live replication floor. Per-query
847 scores correlate across regimes at $r=0.90$ – 0.92 , and the balanced threshold makes the same fire/hold
848 decision as the headless read on 0.887 (clean) and 0.912 (overlap) of queries. The end-of-turn read
849 costs 24–26 ms at p50 ($p95 \leq 30$ ms) in both arms, with the first audible answer chunk arriving ≥ 0.5 s
850 after the gate decision ($p50 \approx 0.58$ s), so the decision precedes any audible commitment. Scope: this
851 loop keeps per-turn session control — the mic stays open while the talker speaks and interruptions
852 seed the next read, but incoming audio is not prefilled *during* generation. In hindsight this sweep
853 and the concurrent arm are best read as *read-point ablations*: the deployed native full-duplex regime
854 (Appendix O) sits strictly between the turn-based ideal and the harness-concurrent worst case on
855 every pool measured. The concurrent regime is measured separately below.

856 **Floor control under escalation: barge-in vs. backchannel.** The demonstration loop’s floor pol-
857 icy (speech over the talker ducks playback; a fast ASR pass returns the floor on backchannels or
858 commits the interrupt) raises a distinct worry: did adding the gate change the talker’s overlap behav-
859 ior? A dedicated sweep injects overlap stimuli into the deployed loop at real-time pacing — four
860 classes: short backchannels (0.40–0.46 s, the English+Chinese lexicon), long lexicon-only contin-
861 uers (time-stretched to 1.33–2.08 s, probing the sustained-speech commit rule), out-of-lexicon stop
862 commands, and pool queries spoken over the answer — in matched pairs with the gate on and off
863 under an identical floor policy (non-firing queries, $n=40$ per class per arm). The answer is exact
864 orthogonality: all 160 pairs make the identical duck/resume/interrupt decision in both arms (every
865 paired rate delta 0.000 [0.000, 0.000]; paired latency medians within 31 ms), down to the single seed-
866 ing failure, which occurs on the same query in both arms. By construction the gate reads once at end
867 of turn and never enters the floor state machine; this measures it. The three phases that exist only
868 under escalation — the stall sentence, the expert wait, the relay speech — are the actual new risk

surface and behave (Table 13): a genuine barge-in aborts the escalated turn and seeds the next turn 40/40, and short backchannels hold the floor at the same rate as during local answers. One asymmetry is disclosed: a false interrupt during the expert wait cancels the pending expert call, so the policy’s fail-toward-interrupt bias is recoverable but not free there. That residual backchannel false-interrupt rate (0.40 short, 0.50 long; identical in both arms) is a property of the harness policy and is per-stimulus deterministic: non-lexical hums (“mm-hm” and its Chinese counterpart) die by the ASR-empty fail-closed rule (16/16 each; worded tokens such as “okay” 0/16), and continuers sustaining loud speech past the ≥ 1.2 s commit rule die without any ASR check (20/20; pause-bearing variants 0/20). We keep the policy as deployed for measurement integrity and note both rules as the obvious fixes. For scope: the checkpoint’s *native* duplex profile, measured separately on Full-Duplex-Bench v1.0 [Lin et al., 2025] (727 samples, official scripts), shows best-in-class pause handling (turn-over rate 0.125/0.117 Candor/synthetic), competitive user-interruption response (0.915 at 0.90 s), and *zero* spontaneous backchannels — the lexicon floor is a harness-level approximation added precisely because the talker head has no native backchannel behavior to preserve.

overlap injected during	backchannel holds floor	barge-in aborts	seeds next turn
local answer (gate off)	0.60 [0.45, 0.75]	40/40	39/40
local answer (gate on)	0.60 [0.45, 0.75]	40/40	39/40
stall sentence	0.63 [0.38, 0.88]	16/16	16/16
expert wait	0.60 [0.33, 0.87]	14/14	14/14
relay speech	0.55 [0.27, 0.82]	10/10	10/10

Table 13: Floor-control sweep over the deployed voice loop. Short backchannels (lexicon bursts) vs. genuine barge-ins (pool queries spoken over the talker), by the phase the overlap lands in; the last three phases exist only under escalation. Interrupt latency p50 1.39–1.42 s from overlap onset in every phase; escalated-turn aborts discard the pending expert answer and the barge-in speech becomes the next turn.

Concurrent prefill: interleaved listen/speak in one KV stream. A third arm drives the streaming API in time division: while the talker is mid-answer, each 1 s chunk of the target query is prefilled into the *same* session between generation chunks, so the end-of-turn read fires while generation is still active (238/240 queries, full interleave coverage). The carrier answer is teacher-forced: under sampling the duplex model reliably yields the turn — it emits EOS within ~ 2 s of user audio appearing in its context, the trained turn-taking behavior surfacing under interleave. This is the one shift on the transfer ladder that breaks the frozen read: scores drift up (mean 0.48→0.71; the frozen balanced threshold’s fire rate jumps 37→75%), AUC drops 0.869→0.758 [0.693, 0.816] (paired Δ AUC -0.109 [$-0.158, -0.064$]), and label-free requantiling recovers decision agreement only to 75%. The information, however, survives in the state: refitting the *same* 12288-d linear read on 360 calibration queries collected in-regime reaches **0.818** [0.761, 0.871] on the held-out 240 (calibration OOF 0.775) — more than half the gap recovered at a quarter of the deployed probe’s calibration scale. The mid-layer competence direction persists under concurrent listen/speak; unlike every earlier shift (modality, external pools, talker-on), calibration must follow the regime. Run-to-run replication of the concurrent read correlates at 0.895. Caveat: the interleave goes through the public streaming API — turn-role headers enter mid-generation and the carrier text is forced — so this measures the concurrent *state*, not the official duplex serving format.

The full loop in the concurrent regime. With the in-regime gate (thresholds = label-free quantiles of each pool’s own concurrent never-arm scores), the complete escalation loop — carrier speaking, target audio interleaved, EOT read mid-generation, stall → expert → relay — was re-run end to end on all seven pools (Table 14). We report this sweep as *exploratory*: one session per query per arm, an in-regime probe calibrated on 360 internal rows only, teacher-forced carrier (the caveat above). The loop’s mechanics reproduce — realized escalation rates track nominal on the external pools (10.8–51.2%), and latency holds (EOT read p50 23 ms, canned-stall onset 27–34 ms, local first audio p50 .655 s vs. .552 turn-based, the price of the longer concurrent context) — and on the internal pool, where calibration matches the regime, so does *selection*: accuracy

909 climbs 40.4→52.1→66.2% (speakable **44.5→56.0→71.1%**) over the balanced and aggressive tiers,
 910 clearing the matched-rate random remix of the same never/always outcomes ($p=.017$ and 5×10^{-5} ;
 911 speakable .009 and 5×10^{-5}), and *selective escalation beats always-escalate* (66.2 vs. 61.7% at two
 912 thirds of the calls; speakable 71.1 vs. 66.5) — the turn-based signature result carries over. The
 913 matched-random rows of Table 14 bound any stronger reading, in three ways. (i) Gains are not
 914 monotone everywhere: the internal conservative tier under-fires (2.5% realized vs. 15.8% nominal
 915 — the top calibration quantile did not transfer to the regime; balanced and aggressive did) and lands
 916 flat (40.4→40.0, 44.5→43.6, inside the ± 2 –3 floor). (ii) Externally the 360-row in-regime probe
 917 separates from matched-rate random only in spots (Llama @50%: 90.8 vs. 86.5, $p=3\times 10^{-4}$; Rea-
 918 soning zh @30/50%: $p \leq .0024$; AlpacaEval @15%: $p=.031$); at the other external operating
 919 points the budget gains are what random escalation would buy — consistent with this probe’s external
 920 AUC of .57–.67 against the turn-based probe’s .68–.81 (internally, where calibration matches the
 921 distribution, it reads .857 vs. the turn-based .879). We then tested the calibration-scale attribution di-
 922 rectly: the full 2,310-row calibration mix was re-collected in the concurrent state and the same read
 923 refit at full scale. Scale is real — every English pool lifts (paired against the 360-row probe on the
 924 same features: TriviaQA .645→.699, WebQ .641→.713, Llama Q. .674→.755, SD-QA .639→.701;
 925 mean $\Delta\text{AUC} +.068$ [.036, .099]) along a monotone data-scaling curve (external mean .625→.689
 926 over 360→2,310 rows; internal .818 unchanged) — but it buys only about a third of the distance to
 927 the turn-based .785–.806, and Reasoning zh, absent from the English calibration mix, moves within
 928 noise (.615→.577 [−.117, .037]). The residue is the regime itself, not calibration scale. Table 14’s
 929 loop ran with the 360-row probe and its numbers are unchanged. (iii) On Llama Questions selective-
 930 vs-always is judge- and noise-sensitive (official judge 90.8 vs. 91.2; ours 86.4 vs. 85.6 — both deltas
 931 inside the floor), so we claim that comparison only on the internal pool. Floors move by pool — the
 932 regime cost story: Llama $\approx$ unchanged, TriviaQA −16, Reasoning zh −18 (the English carrier hurts
 933 the zh pool most), WebQ +4.8 under the official judge (within the ± 2 –3 replication floor; the larger
 934 our-judge shift is alias-matching style sensitivity). AlpacaEval rises 4.36→4.60 (1–5 scale, always
 935 4.76), but only its conservative tier clears matched random ($p=.031$; 4.40–4.55 across tiers), so the
 936 turn-based honest-negative stands.

Table 14: The full escalation loop in the concurrent regime (*exploratory*: one session per arm, 360-row in-regime calibration, teacher-forced carrier), all pools, judge-matched to Table 1 (OAB official for TriviaQA/WebQ/Llama; ours for SD-QA/Reasoning/internal; VoiceBench 1–5 for AlpacaEval). Gate = in-regime probe, per-pool label-free quantile thresholds; tier columns are headed by each pool’s *realized* rate, and the *rand* row under each pool is the matched-rate random remix of the same never/always outcomes (20k draws; $p = P(\text{random} \geq \text{gate})$, marked * $p < .05$, ** $p < .01$). Accuracies in %; AUC = in-regime probe vs. never-arm failure, decimals. Script: scripts/20_conclive_random_check.py.

pool	never	cons.	bal.	aggr.	always	AUC
TriviaQA	50.4	54.0	64.0	71.6	91.6	0.636
<i>rand</i> (10.8/30.8/46.8%)		54.9	63.1	69.6		
WebQ	61.6	65.2	68.8	71.2	81.6	0.636
<i>rand</i> (15.6/30.4/48.8%)		64.7	67.7	71.4		
Llama Q.	81.6	83.2	84.8	90.8**	91.2	0.672
<i>rand</i> (14.8/29.2/51.2%)		83.0	84.4	86.5		
SD-QA	49.5	55.5	60.5	70.5	89.5	0.636
<i>rand</i> (15.0/29.5/50.5%)		55.5	61.3	69.7		
Reason. zh	40.6	48.5	58.4**	64.4**	80.7	0.570
<i>rand</i> (15.3/30.2/45.0%)		46.8	52.7	58.7		
internal (240)	40.4	40.0	52.1*	66.2**	61.7	—
<i>rand</i> (2.5/36.2/66.2%)		41.0	48.1	54.5		
internal speakable (218)	44.5	43.6	56.0**	71.1**	66.5	0.857
<i>rand</i> (2.3/31.2/62.8%)		45.0	51.4	58.3		
AlpacaEval (1–5)	4.36	4.45*	4.53	4.60	4.76	—
<i>rand</i> (10.1/30.7/47.7%)		4.40	4.48	4.55		

937 **A serendipitous recovery behavior.** An accidental run surfaced one bonus behavior worth report-
 938 ing: when a long question contains a >1.25 s internal pause, the endpointer fires early and the talker
 939 starts answering the fragment — but the question’s own continuation then triggers barge-in, and
 940 the remainder becomes the next turn, where the gate simply re-scores it. Early end-pointing thus
 941 degrades into an extra turn rather than a lost query — the honest duplex-ish recovery available to a
 942 turn-based loop.

943 I Fixed-threshold transfer and the expert-benefit oracle

944 Both analyses below run in the *gold-inject* view (escalated rows take the expert’s answer to the gold
 945 text). The deployed-channel view cannot evaluate a freely moving threshold: the most permissive
 946 arm escalates 50%, so the other half of each pool has no expert outcome on record, whereas the
 947 ceiling runs cover every id. Labels follow each pool’s own judge (Table 1); the two judges disagree
 948 by up to 13 points on Web Questions, so they are never mixed within a row. AUC columns are
 949 recomputed from the deployed sweep’s stored end-of-turn scores; they differ from Table 1’s AUC
 950 column — captured in a separate scoring pass — by up to 0.03 externally and 0.04 on the internal
 951 pool (0.840 here vs. 0.879 there): the passes render and prefill the audio independently, and the
 952 probe score carries render-to-render jitter. AlpacaEval is excluded — it has no binary correctness
 953 label on either side, only a 1–5 score whose expert ceiling is degenerate (195/199 at 5).

954 **One absolute threshold, six pools.** The deployed gate adapts per pool, but only through label-free
 955 quantiles of its own score distribution. To separate what the probe’s *ranking* carries from what its
 956 *scale* carries, we freeze the single absolute threshold the internal system actually deployed ($\tau=0.628$,
 957 the balanced tier) and apply that number verbatim everywhere. **Scale does not transfer:** the same
 958 τ realizes escalation rates from 2.0% (Reasoning QA) to 48.0% (SD-QA), a 24-fold spread, so a
 959 fixed cut cannot hold an escalation budget across pools. **Ranking does:** at whatever rate it fires,
 960 fixed- τ escalation beats a random reference at the same realized rate on four of five external pools,
 961 paired-bootstrap interval excluding zero (Table 15). The exception is Reasoning QA, where τ fires
 962 on 2% of queries and there is almost nothing to measure. This is the concrete case for the label-free
 963 quantile step: it buys budget control, not discrimination.

964 **Does the gate find failures the expert can fix?** The probe is fit and read against *local failure*, but
 965 what a routing system needs is *expert benefit*: $y=1$ when the local model is wrong *and* the expert is
 966 right. Re-scoring the same frozen scores against this harder label costs little on the retrieval pools
 967 (Speech TriviaQA 0.796 $\rightarrow$ 0.782, Llama Questions 0.830 $\rightarrow$ 0.813) and more where the expert is it-
 968 self weak (Web Questions 0.773 $\rightarrow$ 0.689 against a 0.864 expert ceiling; internal 0.840 $\rightarrow$ 0.732). The
 969 gate thus remains a usable benefit predictor out of distribution (0.63–0.81) despite never having seen
 970 an expert outcome, and the drop is largest exactly where the paper already reports a flat live curve.
 971 On Web Questions the mechanism is direct: 75.0% of local failures are expert-fixable pool-wide, but
 972 only 65.5% of the failures inside the gate’s top-scoring 15% are, and that share climbs monotonically
 973 as the cut widens (0.655/0.685/0.711 at 15/30/50%) — the gate’s most confident failure calls are the
 974 ones the expert also misses. A benefit-*trained* refit closes none of this: with expert outcomes col-
 975 lected for all 2,310 calibration queries (gold-text expert, expert-right rate .853), retraining the same
 976 12,288-d read directly on the benefit label moves internal benefit-AUC .732 $\rightarrow$.759 (paired +.027
 977 [.004, .052]) and leaves every external pool inside noise (Δ $-.029$ to $+.037$, all CIs spanning zero),
 978 at no internal cost on the fail label (.840 $\rightarrow$.839). The benefit gap reflects the label’s dependence
 979 on the expert’s own failure surface, not a trainable objective mismatch — the expert-agnostic label
 980 leaves nothing on the table. Table 16 places the deployed gate against the matched-budget oracle
 981 that escalates true benefit cases first: the gate clears random at every pool and budget, capturing a
 982 fifth to three fifths of the random-to-oracle span, and the residual gap is the headroom a decode-time
 983 or expert-aware second stage would have to claim.

Table 15: One absolute threshold ($\tau=0.628$, frozen on the internal pool) applied verbatim to every pool; gold-inject view. Δ is fixed- τ accuracy minus a random reference at the *same realized rate* (paired bootstrap over ids, $B=10^4$, seed 42). AUC columns re-score the same frozen scores against local failure and against expert benefit ($y = \text{local wrong} \wedge \text{expert right}$, base rate in the last column). Rates and accuracies in %; AUC in decimals.

pool	n	rate	local	acc	rand	Δ 95% CI	AUC _{fail}	AUC _{benefit}	base
Speech TriviaQA	250	28.4	66.4	83.2	75.0	[+4.4, +12.1]	0.796	0.782	30.8
Speech Web Questions	250	36.0	56.8	72.0	67.5	[+0.8, +8.4]	0.773	0.689	32.4
Llama Questions	250	8.0	83.6	86.8	84.3	[+0.6, +4.6]	0.830	0.813	11.2
SD-QA	200	48.0	51.5	80.5	71.4	[+4.5, +13.6]	0.800	0.765	43.5
Reasoning QA (zh)	202	2.0	58.9	60.9	59.5	[−0.1, +3.5]	0.679	0.634	30.2
internal frozen test	240	35.0	38.3	65.4	57.0	[+3.8, +13.1]	0.840	0.732	53.3

Table 16: Oracle frontier at matched escalation budgets (gold-inject view): random reference / deployed quantile gate / benefit oracle, which escalates the true $y=1$ cases first. The gate clears random everywhere; the oracle column is the remaining headroom. Accuracies in %.

pool	15% budget			30% budget			50% budget		
	rand	gate	oracle	rand	gate	oracle	rand	gate	oracle
Speech TriviaQA	71.0	76.0	81.6	75.5	83.6	96.4	81.6	91.2	97.2
Speech Web Questions	61.2	64.0	72.0	65.7	70.0	86.8	71.6	76.4	89.2
Llama Questions	85.0	88.8	94.8	86.4	91.2	94.8	88.2	93.2	94.8
SD-QA	57.7	63.5	66.5	63.9	74.5	81.5	72.2	82.0	95.0
Reasoning QA (zh)	63.1	67.8	73.8	67.4	72.8	89.1	73.0	76.7	89.1
internal frozen test	46.3	49.2	53.3	54.3	60.8	68.3	65.0	74.2	88.3

984 J Transfer latency shapes

985 **Matched-rate precision.** Precision at a fixed escalation rate is capped at base-fail/rate: on Llama
986 Questions (base fail 0.16) the 50% cap is 0.32 and the deployed probe reaches 0.304 — 95% of the
987 cap (recall 0.93). Quoting raw precision would misread the strongest pool as the weakest.

988 **The median left-fold.** On easy pools per-arm P50 is non-monotonic (Llama Questions
989 1.52→1.17→1.60/1.42 s): a verified median-of-a-mixture mechanism, not noise. The probe pref-
990 erentially escalates the queries whose *local* decode is longest — on retrieval pools wrong answers
991 hedge at length (decode time and answer length $r=0.97$), so a wrongness-selecting probe strips the
992 slow tail as a side effect (escalated-at-15% ids: never-arm local P50 1.711 s vs. 1.198 s kept; local
993 accuracy 0.32 vs. 0.73). Controls: the probe does not key on length ($\text{corr}(\text{score}, \text{length}) = +0.05$;
994 ≈ 0 conditioned on wrongness), and on SD-QA — where wrong answers are short — the same probe
995 selects wrongness identically and produces no fold. The mean is monotonic (1.65→2.06 s); kept-
996 local P50 falls 1.52→0.43 s across tiers (Figure 15). On Reasoning QA the shape inverts usefully:
997 the local chain-of-thought is *spoken*, so every reasoning token enters the response clock, while the
998 expert’s chain is hidden behind a flat ~ 3.7 s round-trip — escalation truncates the latency tail (P90
999 13.25→10.94 s) rather than fattening it.

1000 **AlpacaEval scale caveats.** The official 4.8 is an offline chat-mode number — our chat-mode
1001 control reproduces it at 4.86 — and the chat-vs-live gap (4.86 vs. 3.99) is the duplex loop’s spoken-
1002 reply style plus a token cap, so only live arms are comparable to each other. Selected queries score
1003 3.90 on the never arm vs. 3.98 for skipped: no discrimination; escalated rows gain (3.90→4.71)
1004 because the expert writes better long-form answers — random picks would harvest the same gain.

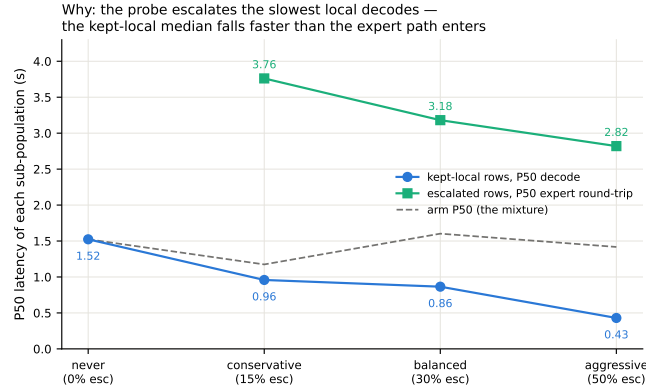


Figure 15: Mixture decomposition of Llama Questions latency per arm: kept-local P50 falls 1.52→0.43 s as the probe strips the slowest local decodes while escalated rows pay a flat ~ 3 s expert round-trip; the arm median (dashed) is their mixture.

1005 K Full-pool and per-family figures

1006 Figures 16 (dual-view) on the full frozen pool, and the complete external-family figure set (Ap-
 1007 pendix L); the noise audit behind the ± 0.02 –0.03 floor is Figure 17.

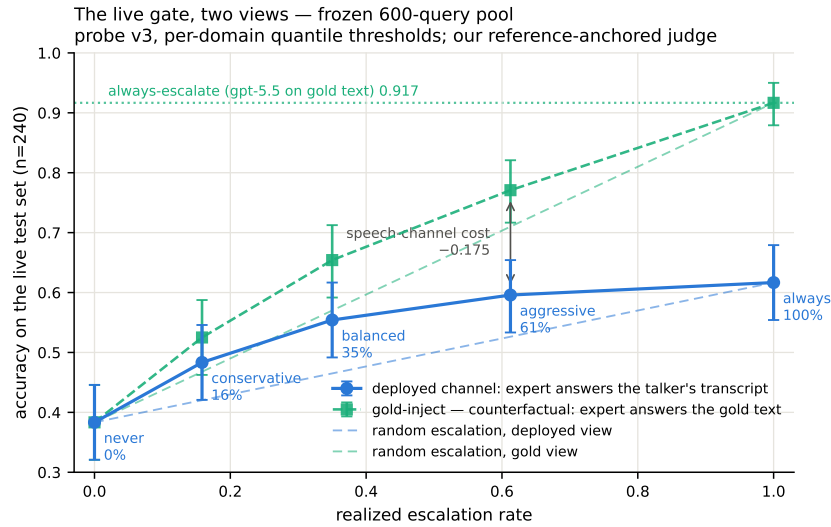


Figure 16: Dual-view live tradeoff, full pool ($n=240$); companion to Figure 12.

1008 L External-benchmark figure set

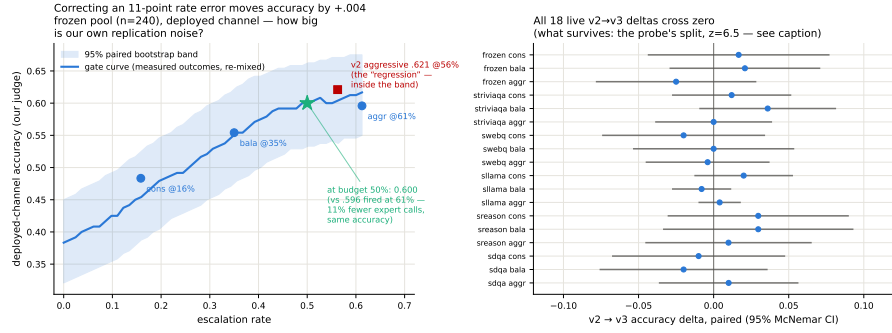


Figure 17: Replication-noise audit: same query, same audio, re-run across arms. Kept-local verdicts flip 2.3–18.8% by pool; repeated escalation 0.7–10.6%; implied paired SE 0.009–0.028.

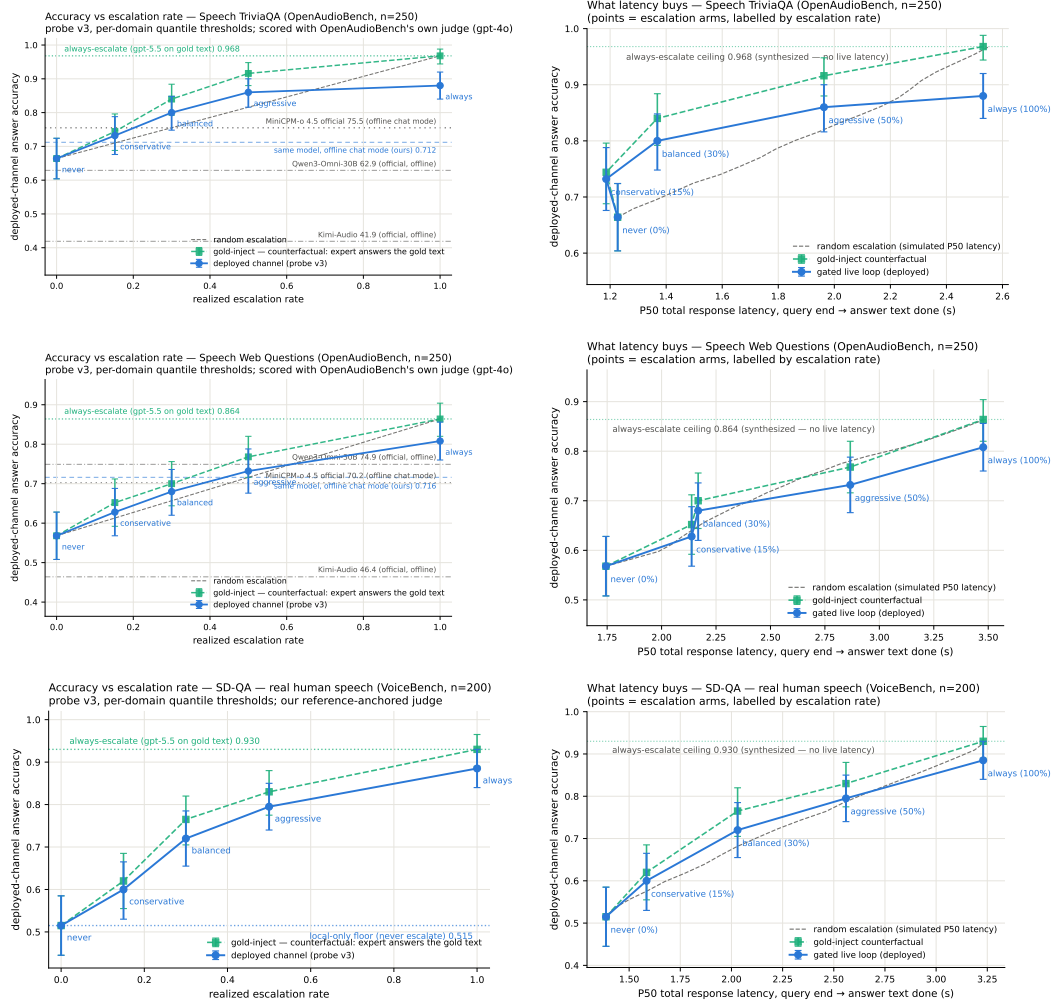


Figure 18: Speech TriviaQA, Speech Web Questions, SD-QA: dual-view (left) and accuracy-latency (right).

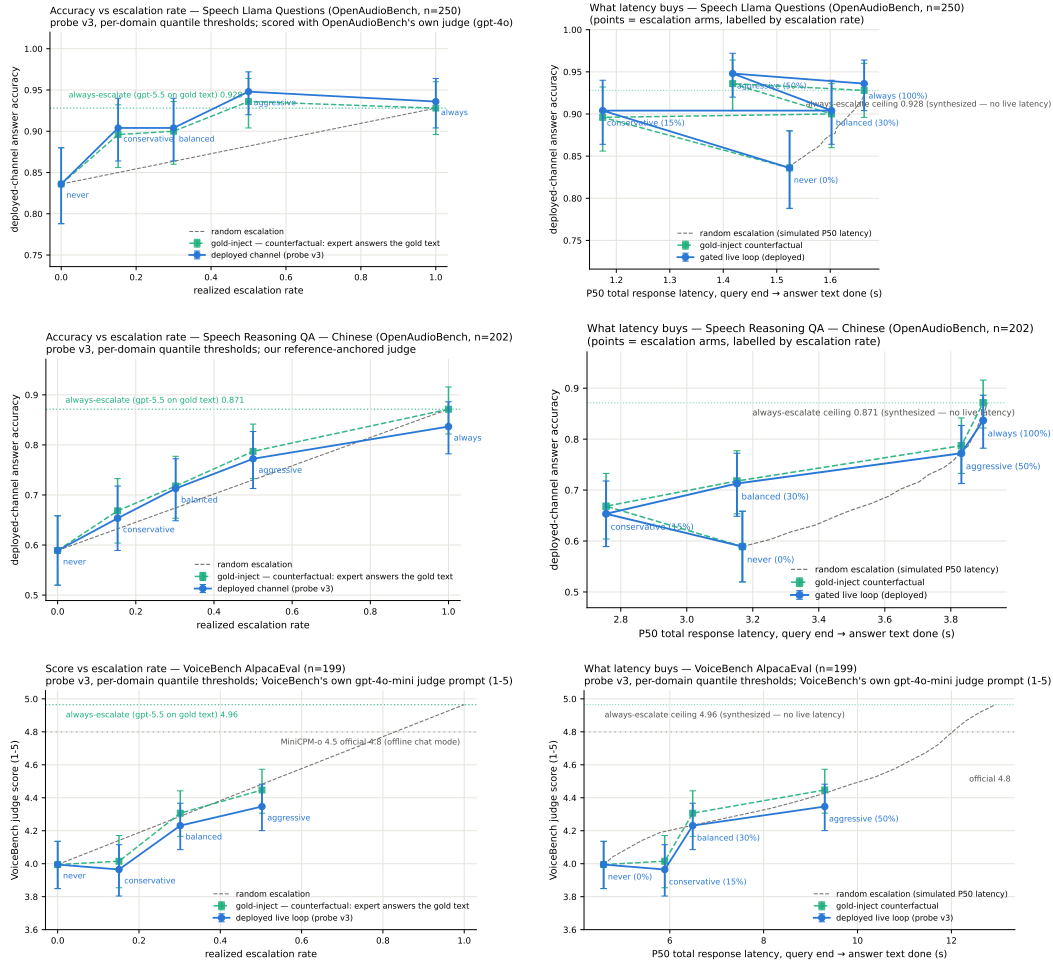


Figure 19: Llama Questions, Reasoning QA (zh), VoiceBench AlpacaEval: dual-view (left) and accuracy–latency (right).

1009 M Judge and self-evaluation prompts

1010 All adequacy labels use the following fixed system prompt (structured outputs with a strict {reason,
1011 adequate} schema; the judge never sees which system produced the answer):

```
1012     You are a strict, fair grader of a small language model’s answers.  
1013     You are given a QUESTION, an optional REFERENCE answer, and a  
1014     CANDIDATE answer produced by a small model. Decide whether the  
1015     CANDIDATE is ADEQUATE, meaning ALL of: (1) factually correct;  
1016     (2) any reasoning shown is valid; (3) it actually and completely  
1017     answers the question that was asked. For multiple-choice questions  
1018     the candidate must land on the correct option (by letter or by  
1019     unambiguously stating the option’s content). A partially correct,  
1020     hedged, refused, off-topic, or truncated answer is NOT adequate. If  
1021     a REFERENCE is provided, grade against it. Be strict but do not  
1022     penalize harmless differences in format or extra correct detail.
```

1023 The verbalized self-evaluation baselines use one fixed phrasing each; $P(\text{Yes})$ is read from the first
1024 generated token’s distribution:

```
1025     [pre] I am going to show you a question. Do NOT answer it. Judge  
1026     honestly whether you yourself would answer it correctly. Question:  
1027     {q} Would you answer this question correctly? Reply with exactly one  
1028     word: Yes or No.  
1029     [post] Here is a question and a proposed answer. Question: {q}  
1030     Proposed answer: {a} Is the proposed answer correct? Reply with  
1031     exactly one word: Yes or No.
```

1032 N Reproducibility

1033 All sampling and splits use a fixed seed (42); the calibration/test split is frozen once and the test
1034 split is touched only by the evaluation scripts. Environment pins: torch 2.8.0; transformers 4.51.0
1035 for MiniCPM-o 4.5, with per-model pins for comparison backbones; all experiments run on single-
1036 H100 instances. Probe artifacts (weights, feature configuration, per-pool label-free thresholds), the
1037 judge prompt above, all trace files, and the per-experiment execution log — including per-run GPU
1038 and API spend, on the order of \$500 total — ship in the project repository, released upon publication.

1039 O The native full-duplex regime

1040 Every result so far reads the probe under a *turn-based* serving loop (harness-controlled end-of-turn)
1041 or its harness-interleaved concurrent variant (Appendix H). MiniCPM-o 4.5 also ships a native full-
1042 duplex mode: audio prefills in 1 s units into the same context the model generates from, and the
1043 head itself emits ⟨listen⟩ or speaks, yielding the floor via its turn-end token. This is the deployed
1044 regime of the demonstration system, and this section recalibrates and re-validates the gate inside
1045 it. The read point needs no external end-of-turn detector: the chunk where the head first switches
1046 listen→speak *is* the commit-to-answer moment, and the probe reads there (the tail- k features then
1047 cover the model’s first ~ 8 answer tokens — a generation-side read the head hands us for free). For
1048 87% of test queries this commit lands within ± 1 chunk (± 1 s) of the true question end.

1049 **Recalibration.** The full 2,310-row calibration mix was re-collected in this regime (fresh duplex
1050 session per query, features at the deployed read point, labels unchanged) and the same 12,288-d
1051 linear read refit. On native test-240 features the in-regime probe reaches AUC **0.830** [0.777, 0.878]
1052 (transferred concurrent probes: 0.807–0.818; turn-based reference 0.877). Externally the in-regime
1053 refit beats both transferred probes on all five pools — TriviaQA .711 ($\Delta +.080$ [$+.030, +.129$]), WebQ
1054 .736, Llama-QA .757 ($\Delta +.118$), SD-QA .736, Reasoning-zh .606 ($\Delta +.115$); external mean .709.
1055 The regime ordering is monotone everywhere: turn-based (.877/.771 internal/external mean) > na-
1056 tive (.830/.709) > concurrent (.689 external) — i.e. a substantial share of the concurrent regime’s

Table 17: Native-regime probe AUC at the deployed listen→speak read point, under the OFFICIAL serving configuration (final deployed system; the chronological narrative above measured under inherited wrapper defaults). The merged refit adds the 2,700-row public-benchmark expansion (incl. 400 Mandarin rows) to the official core. En-4 mean .770 vs. turn-based .771 — the English regime cost is eliminated; the Mandarin slice restores Reasoning-zh from the all-English chance collapse.

pool	<i>n</i>	old-config	official (2,542)	official+exp3 (5,228)
internal test	239	.830	.846	.844
TriviaQA	250	.711	.759	.767
WebQ	241	.736	.754	.772
Llama-QA	245	.757	.789	.815
SD-QA	200	.736	.737	.728
Reasoning-zh	202	.606	.495	.605

Table 18: Native-regime gated accuracy under the official serving config and the deployed merged (official+exp3) gate (8ad remix: native scores select, native local outcomes + cached always-arm expert outcomes mix). *p* = permutation vs. matched-rate random; escalation rate in parentheses. On frozen test the aggressive tier reaches within .012 of the always-escalate ceiling at 48% of the expert cost. Reasoning-zh now separates (AUC .605) but global English thresholds still barely fire there — the per-pool quantile remedy applies. AlpacaEval (measured under the 2,542-row official gate) stays a clean honest negative.

pool	local	balanced	aggressive	always
frozen test	.431	.556 (23%, <i>p</i> <.0001)	.657 (48%, <i>p</i> <.0001)	.669
TriviaQA	.568	.704 (20%, <i>p</i> <.0001)	.852 (61%, <i>p</i> <.0001)	.916
WebQ	.411	.560 (29%, <i>p</i> =.025)	.734 (65%, <i>p</i> <.0001)	.809
Llama-QA	.767	.800 (5%, <i>p</i> =.0005)	.865 (29%, <i>p</i> <.0001)	.910
SD-QA	.455	.620 (27%, <i>p</i> =.0055)	.810 (68%, <i>p</i> =.001)	.895
Reasoning-zh	.525	— (1%)	.574 (13%, n.s.)	.807
AlpacaEval (VB 1–5)	3.65	3.70 (6%, n.s.)	4.07 (37%, <i>p</i> =.24)	4.76

1057 cost was the harness read point, not liveness itself. Thresholds again do *not* transfer (balanced
1058 quantile 0.386→0.645); scores are decoding-temperature invariant (0.7-trained probe on temp-0.1
1059 features: AUC 0.843).

1060 **Validity.** Re-mixing native local outcomes (gpt-judged, deployed decoding) with cached always-
1061 arm expert outcomes under the native scores (Table 18): under the deployed merged gate, gated accu-
1062 racy clears matched-rate random on all five English pools at both deployed tiers — frozen .431→.657
1063 at 48% escalation (*p*<.0001; ceiling .669), TriviaQA .568→.852 at 61%, WebQ .411→.734, Llama-
1064 QA .767→.865, SD-QA .455→.810 at 68%. In-regime training renormalizes the score distribution,
1065 so realized fire rates sit near their nominal quantiles on the English pools (the earlier 64–73% drift
1066 was a transferred-threshold artifact). On Reasoning-zh the gate simply never fires: the all-English
1067 calibration places zh scores below every global threshold — the language axis of Appendix H, now
1068 expressed as a fire-rate failure rather than an AUC failure. Both operating-point failures are fixable
1069 without labels: thresholding each pool’s own recent-score quantile (the converged limit of a 100-turn
1070 sliding-window quantile tracker — no labels, no pool identity) realizes the nominal fire rate on every
1071 pool including Reasoning-zh. The talker’s own local floor is lower in-regime (frozen .483→.371;
1072 ≈one third decoding temperature, ≈8 points genuine regime cost), which *raises* the gate’s marginal
1073 value in deployment. The in-regime calibration scaling curve mirrors the concurrent one and is not
1074 saturated: internal .717→.830, external mean .643→.709 from 360→2,310, and a further public-
1075 benchmark expansion to 5,009 rows (2,300 English across the same seven families + 400 Mandarin:
1076 MGSM-zh and XCOPA-zh through the same TTS pipeline, source-disjoint from Reasoning-zh) con-
1077 tinues the ≈+.02-per-doubling external trend to **.736** while internal stays flat (.834) — and the
1078 Mandarin slice moves the axis English data never touched, Reasoning-zh .606→.659 (Δ vs. conc-
1079 2310 +.169 [+0.085,+.251]).

1080 **Live validation.** Full live runs on frozen test-240 (real raw-audio ASR uplink, real expert, wait
 1081 paced at one chunk per second) validate the remix arithmetic where the channel is thin and expose
 1082 where it is not: balanced delivers .504 at 23% fire (remix predicted .492 at 24%); aggressive delivers
 1083 .537 at 45% against a .596 remix prediction, and the gap decomposes entirely onto fired turns —
 1084 live fired-turn accuracy .495–.589 vs. the .667 cached-expert ceiling. The native *relay channel* (the
 1085 answer re-entering as a text unit and being voiced) is the regime’s lossy element — 89% of relay
 1086 units need the nudge retry — playing the role the transcription uplink played in the turn-based loop;
 1087 the uplink itself is now clean (raw-audio ASR, p50 0.9 s). Latency, expert-cache warm: expert p50
 1088 4.4–5.7 s, wait p50 5–6 chunks, fire→relay first audio p50 ~7–8 s. Selection is visible live: unfired
 1089 subsets score .478/.573 against the .371 unconditional local floor.

1090 **Dialogue-act gating.** Full duplex creates commits that are not answers: after a barge-in or
 1091 a backchannel the head speaks to *manage the floor* (“Okay.”), and the failure probe is out-of-
 1092 distribution there — on 194 TTS’d floor-management stims (stop commands, backchannels, ac-
 1093 knowledgments, fillers; en+zh) the unmodified gate false-fires on 37% at the aggressive tier (stop
 1094 commands 45%), i.e. it would escalate “stop” to the expert. The fix is a second linear head on
 1095 the *same* L22 read — info-seeking vs. floor-management — which separates the two acts perfectly
 1096 (OOF AUC 1.000); thresholded at the question-side 0.5th percentile it costs 0.5% of true escalations
 1097 and admits 0% of floor stims. Escalation requires *both* heads: $\text{act} \geq \tau_a \wedge P(\text{fail}) \geq \tau$. The de-
 1098 ployed demo answers floor turns naturally and reports the act score per commit; the failure probe’s
 1099 calibration is untouched.

1100 Live use then sharpened both heads within minutes — scripted pools miss what ten minutes of real
 1101 conversation finds. (a) Live mic speech shifts both act distributions into the calibration gap (real
 1102 request-phrased questions score ~.93, live floor turns ~.37 vs. a standalone-stim maximum of .05):
 1103 a percentile-anchored τ_a rejected a real question and the talker delivered an empty promise. The fix
 1104 is a gap-center threshold plus *in-context* calibration stims — each utterance dumped as the second
 1105 turn of a carrier conversation, matching how floor acts actually arrive; the act read stays perfectly
 1106 separable (AUC 1.000) under both conditions, and in-context stop commands false-fire the *failure*
 1107 probe on 57% at the aggressive tier, so the act term is not optional. (b) The native refit had silently
 1108 dropped the FreshQA real-time extension (“the stock price of NVDA today” scored $P(\text{fail})=.29$
 1109 and stayed local); replaying that recipe in-regime restores fast-changing heldout fire to .45/.75/1.00
 1110 across tiers at zero internal-AUC cost (.830→.833), with budgets still quantiled on the core mix. (c)
 1111 A relay’s own speak-onset must not re-enter the gate (re-fire with an empty uplink); deliveries are
 1112 guarded to their end of turn. The pattern across all three: the mid-layer signals separate cleanly
 1113 every time, and what re-opens at each regime or domain shift is calibration coverage and thresholds
 1114 — interactive use is itself an evaluation modality.

1115 **Multi-turn grounding.** A stateless expert uplink breaks on follow-ups: “what is NVDA trading
 1116 at” then “what about Apple” sends only the second utterance, unresolvable in isolation. The probe
 1117 itself is unaffected — it reads L22 with the prior turns live in the model’s KV cache — so the fix is
 1118 in the uplink, not the gate: the deployed demo threads a rolling resolved-dialogue history into the
 1119 expert input and asks it to resolve references, after which the follow-up correctly returns Apple’s
 1120 share price. The measured eval pools are single-turn, so this is a deployment capability rather than
 1121 a table; systematic multi-turn evaluation is future work.

1122 **Floor control, natively.** With the harness gone, 8ba’s question re-poses on the head itself (108
 1123 scripted overlap trials, stims spoken over the talker at fixed offsets). During answer speech,
 1124 backchannels almost never stop the floor (false-stop 3/24 within a 6-chunk window) with *no* ASR
 1125 or lexicon anywhere in the loop; sustained question speech yields the floor (6/12 within 6 chunks,
 1126 median 8.5 post-stim); short burst commands (“Stop!”) mostly do not (3/12) — the native head
 1127 reads sustained speech, not commands, a naturalness-for-reliability trade the old energy-VAD did
 1128 not make. The escalation machinery is floor-transparent where it should be: relays complete un-
 1129 der overlapping stims (26/30), and backchannels during the thinker wait leave the pending relay
 1130 intact (8/10). The weak stop-command sensitivity decomposed into two causes, found in sequence.

1131 *Mechanism*: instrumenting the serving loop shows the head never samples $\langle \text{listen} \rangle$ mid-turn — zero
1132 attempts across 36 trials — so its only stop channel is the turn-end token. *Configuration*: a user-
1133 demanded vanilla control (the bare official serving stack, no gate machinery) revealed that our loop
1134 had inherited the checkpoint wrapper’s sampling defaults ($k=100$) rather than the official demo’s
1135 serving config ($k=20$, a 3-chunk startup listen guard, an assistant-style system prompt). Under the
1136 official config the turn-end channel is *responsive*: a spoken “stop” two seconds into an enumeration
1137 answer ends the turn in 2.0–2.2 s in 3/3 scripted sessions, versus 1–13 s (once running the answer to
1138 completion) under the inherited defaults — wide- k sampling dilutes away the rising turn-end proba-
1139 bility. The deployed demo now serves the official config, and the earlier floor-control stop numbers
1140 measured under the inherited defaults should be read as a lower bound on native responsiveness. Two
1141 durable lessons: serving-parameter parity with the reference stack is part of “native”, and a vanilla
1142 control arm belongs in the harness from day one. The one true interaction: a *new question* during
1143 the wait derails the pending relay 7/10 times — the head simply moves on — which is the empirical
1144 case for aborting the in-flight expert call on user speech, left as deployment future work. Latency:
1145 listen chunks cost 0.04 s and speak chunks ~ 0.5 s on one H100 (realtime holds); fire→stall-audio
1146 0.2–1 s; fire→relay-first-audio is the expert wall clock (13–20 s observed) plus ~ 2.5 s.